Pergamon

Vistas in Astronomy, Vol. 36, pp. 311–361, 1993
Printed in Great Britain.
0083-6656/93 $24.00

0098-2997(93)E0003-7

ASTRONOMY AND THE LIMITS OF VISION

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ABSTRACT

Celestial visibility is the study of the limits of observability of objects in the sky, with application to deducing the truth about historical events or to the derivation of astronomical information of modern utility. This study is based on what is seen by ordinary humans, either in their everyday lives or at times of historical events. The results of such studies have more relevance to non-scientists than does any other area of astronomy. Celestial visibility is a young discipline in the sense that the number of interesting applications with simple solutions outnumber the solved problems; it is a broad interdisciplinary field that involves work with astronomy, meteorology, optics, physics, physiology, history, and archeology. Each of these disciplines contribute specialized mathematical formulations which quantify the many processes that affect light as it leaves a source, traverses the atmosphere, and is detected by the human eye. These formulas can then be combined as appropriate to create mathematical models for the visibility of the source under the conditions of interest. These model results can then be applied a wide variety of problems arising in history, astronomy, archeology, meteorological optics, and archeoastronomy. This review also presents a dozen suggestions for observing projects, many of which can be directly taken for individual study, for classroom projects, or for professional research.

1. Introduction

Celestial visibility is an interdisciplinary research field concerned with what can and cannot be viewed in the sky by visual observations, and is relevant to solving many mysteries and phenomena from history and astronomy. In the days before the invention of the telescope, all astronomy was done with the unaided eye. In both ancient and modern times, historical events have been influenced by visual observations of sky events. There are many sights in the sky for which valid explanations have become available only recently. For studies of ancient chronology as well as for calendar-making even to the present day, studies of lunar and planetary visibility are vital. Therefore, if we seek to understand old astronomical data, the effects of celestial happenings on human affairs, what untrained observers have seen in the sky, or the details of any calendar, then we must learn about "celestial visibility."

By the nature of the topics covered, celestial visibility is concerned with the aspects of astronomy that have relevance to ordinary people. For example, a quarter of the world's current

population sets its festivals and fast periods by the visibility of the thin crescent moon and times its daily prayers on astronomical events. The ancient Egyptians, Mayans, and Aztecs based their calendars and festivals on the heliacal risings of Sirius, Venus, and the Pleiades. Neolithic man apparently constructed numerous temples which were designed with cognoscence of the rising and setting of the Sun, Moon, and stars. The timing of most of the great battles and invasions for at least the last century have been determined in part by lunar phase. Surely most humans have stood out under a clear sky and wondered about the twinkling of stars, the beauty of a sunset, and the glory of a rainbow.

The study of celestial visibility directly relates to many historical questions that fire the public imagination. For example, the results presented below are important to the determination of whether Admiral Peary reached the North Pole, where was Columbus' landfall, whether Stonehenge was an "ancient observatory," how Paul Revere slipped by the British warship in Boston Harbor, what was the Star of Bethlehem, and when was Jesus crucified? Phenomena in the sky have affected major events, ranging from the appearance of a big sunspot provoking a general amnesty in ancient China, to an aurora saving Byzantium from Philip of Macedon, to a lunar eclipse breaking the spirit of the defenders of Constantinople in 1453, to a solar eclipse stopping the war between the Median and Lydian empires. Examples where celestial visibility sheds light on historical events are legion and spread throughout the entire world and through all times.

A detailed knowledge of celestial visibility is required to extract the maximum information out of visual astronomical observations. Thus, extensive effort has gone into the analysis of ancient eclipse reports as the primary record on the accelerations of the Earth's rotation and the Moon's revolution. Exhaustive studies have also been performed on the two millennia of oriental sunspot reports as the primary measure of the long term solar activity. Another major research program has been the analysis of historical supernovas to extract ages, positions, light curves, classifications, peak brightnesses, and rates.

Celestial visibility is both a very old and a very young discipline. The ancient Greeks were constructing models for heliacal rise dates, atmospheric refraction, and eclipse predictions, while medieval Islamic astronomers were much concerned with predicting the first crescent of each month and the times of twilight. While these efforts resulted in empirical rules-of-thumb that usually produced reasonable predictions, the quality of the ancient methods is well below modern standards. Until modern times, the required mathematical tools and physical models were not available, so that no significant advances on the old empirical rules were possible. However, the interests of modern astronomers has been sharply focused on high technology methods (*e.g.*, spectroscopy, radio telescopes, and gamma-ray observatories) and exotic new discoveries (*e.g.*, supernovas, quasars, and gamma-ray bursts). Thus, for the last century, modern astrophysics has passed by questions relating to visual observations of the sky. Nonetheless, research into celestial visibility has been enjoying a renaissance in the last decade, with an ever increasing number of workers in the field. Many issues can be approached for the first time with modern methodologies, so that the number of simple, useful, and startling results is large, just as for a young discipline.

This review article will cover the whole field of celestial visibility. The next section will give detailed equations for most of the phenomena that affect what can be seen in the sky. The middle section presents models of specific visibility questions, of relevance for many applications. The fourth section applies these models to specific questions of historical or astronomical interest. The last section gives twelve suggestions for research projects of current interest.

2. Tools

The theoretical calculation of celestial visibility requires accurate models of many effects. These include models of the source position, refraction, air mass, extinction, source brightness, sky brightness, glare, shadows, resolution, optics, thresholds of human vision, color vision, and weather statistics. In this section, I will present the basic equations and references for each topic. These equations can then be used as tools and combined appropriately for each question regarding celestial visibility.

2.1 Source position

Calculations of celestial visibility almost always begin with an evaluation of the source's position in the sky. In some cases (for example, with the visibility of stars in chimneys or with the green flash), the position can be assumed or low accuracy is adequate. But in many cases, especially those involving historical incidences, the exact position is important. Then, the task can be broken into two parts; first, determining the source's celestial position in right ascension and declination, and second, determining the apparent position with respect to the horizon for the observer.

In general, the most definitive reference for celestial positions of all sources is the *Astronomical Almanac* (or its predecessor the *American Ephemeris and Nautical Almanac*), the *MICA* computer program, and the *Explanatory Supplement to the Astronomical Almanac* (Seidelmann 1992). Unfortunately, these materials are not the most accessible references to many observers and only cover the chronology from the mid-1800's to the near future. A more convenient resource might be any of the many commercial computer programs available for personal computers, of which the *Voyager* program has a good reputation for accuracy. The best reference depends on the objects involved and the needed accuracy. For the Sun, Moon, and planets, the definitive resource is to access the computer tapes of the JPL DE-200 ephemeris or Bretagnon and Simon (1986), while Van Flandern and Pulkkinen (1979), Meeus (1991), Tuckerman (1962), Tuckerman (1964), and Gingerich and Welther (1983) are convenient and suitably accurate for most purposes. For the Moon, the tables of Chapront-Touze and Chapront (1991) are of high accuracy, while Goldstine (1973) and Morrison (1966 and 1968) present times of the lunar phases. For lunar eclipses, the canons of Meeus and Mucke (1979), Espenak (1989), and Liu and Fiala (1992) are convenient and accurate. For solar eclipses, Mucke and Meeus (1983) or Espenak (1987) should be used.

Once the celestial position of the source is known, then the apparent position needs to be deduced. This requires a series of trigonometric calculations which are described in many astronomy

and navigational textbooks (*e.g.*, Smart 1977, Meeus 1991). Note that various corrections might be needed, such as for refraction, parallax, and the difference between Universal Time and Dynamical Time. This last correction is uncertain in ancient eras and may change the local times and visibilities of eclipses and conjunctions by significant amounts (see Stephenson and Morrison 1984 or Meeus 1991).

2.2 Refraction

When light travels through the atmosphere, its path is bent by refraction. Excellent histories of refraction studies are presented by Mahan (1962) and Bruin (1981). For altitudes well above the horizon, the classical treatments of Smart (1977) and Green (1985) are valid. The apparent position of the source is raised by an amount R, the refracted angle, where

$$R = 58.2'' \left[\frac{0.372\,P}{273^\circ + T} \right] \tan Z \qquad \textbf{(1 a)}$$

and where Z is the apparent zenith distance (in units of degrees-of-arc) of the source, P is the atmospheric pressure (in units of millimeters of mercury) and T is the air temperature (in units of degrees Celsius). For altitudes down to the horizon, Saemundsson (1986) gives two alternative formulas for R which are reasonably accurate:

$$R = \frac{1'}{\tan\left(h + \frac{7.31}{h + 4.4} \right)} \qquad \textbf{(1 b)}$$

$$R = \frac{1.02'}{\tan\left(h' + \frac{10.3}{h' + 5.11} \right)} \qquad \textbf{(1 c)}$$

Here h ($= 90^\circ - Z$) is the symbol for the apparent (*i.e.*, as seen) altitude above an ideal horizon, while h' ($= 90^\circ - Z - R$) stands for the true altitude; the units of (h, h') are degrees-of-arc. The amount of refraction on the horizon is typically 34' or 0.57°. For altitudes near and even below the horizon, the complex algorithm of Garfinkel (1967) is highly accurate if the atmospheric thermal structure is similar to that of the U. S. Standard Atmosphere. In Table 1, R-values are presented as determined from independent values h, and as calculated from Equation 1b.

Table 1. Refraction

h (°)	h' (°)	R (')
90	90	0.00
60	59.99	0.57
30	29.97	1.71
20	19.95	2.70
15	14.94	3.64
10	9.91	5.39
5	4.84	9.88
4	3.80	11.7
3	2.76	14.3
2	1.70	18.2
1	0.59	24.3
0	-0.57	34.5

The values of the terrestrial refraction, that is the height of a ray of light emitted horizontally at sea level as taken from the program in Schaefer (1989a), is presented in Table 2. However, Schaefer and Liller (1990) have shown that significant thermal inversions are ubiquitous and will drastically change the refraction close to the horizon. Schaefer (1989a) presents a program for calculating refraction (and air mass) for an arbitrary thermal structure in an atmosphere of any planet.

Table 2. Terrestrial Refraction

Height (meters)	"NORMAL" Distance (km)	With Inversions Distance (km)
0	0	0
16	14.6	21.7
33	21.1	27.3
50	26.1	32.1
67	30.4	36.3
84	34.2	40.1
100	37.6	43.6
150	46.4	53.0
200	53.9	61.0
250	60.4	68.0
300	66.4	74.3
350	71.8	80.0
400	76.9	85.3

2.3 Air Mass

The dimming of a beam of light entering the atmosphere depends on how much air is traversed by the beam. The quantification of dimming can be broken into two components: the first, a geometrical term which is primarily a function of the source's apparent zenith distance (Z), and the second, a meteorological term that varies with time and place (see the next section). The geometrical term (X) is traditionally measured in units of "air masses," which is the optical path length to the source in units of the optical path length towards the zenith.

For altitudes well above the horizon, a good approximation is given by:

$$X = sec(Z) \qquad \text{(2 a)}$$

Near (but not below) the horizon, the formula of Rozenberg (1966) is convenient and fairly good for reasonable elevations above sea level and aerosol density;

$$X = [cos(Z) + 0.025 e^{-11 cos(Z)}]^{-1} \qquad \text{(2 b)}$$

Equations (2 a,b) provide descriptions of the air mass for the atmosphere taken as a whole.

However, the optical path length through the atmosphere has contributions from three sources of extinction: Rayleigh scattering by gas molecules, Mie scattering by aerosols, and absorption by ozone in the stratosphere. Each of these components has a distinct and characteristic dependence on height, so each will contribute a different optical path length for a specific value of Z. The air mass for the Rayleigh and aerosol component can be calculated for the appropriate scale heights with the program of Schaefer (1989a) for arbitrary atmospheric conditions and viewing directions. The ozone air mass can be calculated in a closed form from simple trigonometry.

The air mass for an extinction component that is distributed with an exponential scale height of H_e in kilometers may be calculated with the use of Equation 3a. For Rayleigh scattering, H_e is close to 8.2 kilometers (Allen 1976), while for aerosol scattering the scale height varies substantially with a typical value of 1.5 kilometers (Hayes and Latham 1975). Equation 3b is used to calculate the extinction layer which is H_L above the observer (with R_o being the equatorial radius of the Earth, *i.e.*: R_o = 6378 km), which for the stratospheric ozone is roughly 20 kilometers (Meinel and Meinel 1983). The use of Equations (3 a,b) is vital for any accurate calculation of extinction near the horizon.

$$X_e(H_e) = \left[\cos(Z) + 0.01\sqrt{H_e}\,exp\left(\frac{-30\cos(Z)}{\sqrt{H_e}}\right)\right]^{-1} ,$$

$$X_L(H_L) = \left[1 - \left(\frac{\sin(Z)}{1+(H_L/R_o)}\right)^2\right]^{-0.5} . \qquad \textbf{(3 a,b)}$$

In Table 3 are presented the various air masses as a function of Z.

Table 3. Air Mass

Z (°)	h (°)	sec(Z)	X Eqn(2b)	X_e(8.2 km) (Gases)	X_e(1.5 km) (Aerosols)	X_L(20 km) (Ozone)
0	90	1.0	1.0	1.0	1.0	1.0
60	30	2.0	2.0	2.0	2.0	2.0
80	10	5.8	5.6	5.6	5.7	5.3
85	5	11.5	10.3	10.1	11.3	8.5
87	3	19.1	15.1	14.5	17.9	10.6
88	2	28.7	19.3	18.3	24.9	11.6
89	1	57.3	26.3	24.2	39.3	12.4
90	0	---	40.0	34.9	81.6	12.7

2.4 Extinction

The extinction coefficient (k) is a measure of the total optical path length towards the zenith, and is measured in units of magnitudes per air mass; it is related to the optical depth (τ) as 1.086τ and to

the atmospheric transmission ($T = e^{-\tau}$) as $-2.5 log_{10}(T)$. The extinction depends on both the wavelength of the incident light and the component of extinction being considered.

The Rayleigh scattering component of the extinction is reliably calculated as a function of wavelength, altitude, and the index of refraction by Hayes and Latham (1975):

$$k_R = 113700(n-1)^2 exp[-(H/H_e)]\lambda^{-4} \qquad \textbf{(4 a)}$$

where n is the refractive index of air at sea level for the desired wavelength, λ, in microns (see Allen 1976), and the observer's height is H (in units of kilometers) above sea level. For visual wavelengths ($\lambda = 0.55$ microns), Equation (4 a) is reduced to

$$k_R = 0.1066\, e^{-(H/8.2)} \qquad \textbf{(4 b)}$$

with units of magnitudes per air mass.

The ozone component of the extinction is reliably calculated from the ozone depth from Bower and Ward (1982) and the cross sections in Allen (1976). In the visual bandpass, the ozone depth, D_{oz}, (in mm), and extinction (in magnitudes per air mass) from ozone can be approximated well by

$$D_{oz} = 3.0 + 0.4(\phi\, cos[\alpha_S] - cos[3\phi]) \ ,$$

$$k_{oz} = 0.031 \frac{D_{oz}}{3.0} \qquad \textbf{(5 a,b)}$$

Here, ϕ is the observer's latitude (in radians) and α_S is the right ascension of the Sun. Ozone holes, ozone depletion, biennial cycles, and even major storm systems will all change the ozone depth from the average values given by Equation 5a. Equation 5b applies for the central wavelength of day vision, while the coefficient is only 30% as large for night vision. Since k_{oz} is small compared to other components, its uncertainties have little impact upon the overall extinction problem.

Atmospheric aerosols come from many sources, including sea spray, windborne desert dust, and tree pollen. They vary in complicated manners that are difficult to predict with accuracy. Nevertheless, there are a variety of trends which can be used to provide a reasonable guess for the aerosol extinction coefficient (k_a). From an evaluation of trends for time of year, relative humidity, altitude, and wavelength, I find that

$$k_a = 0.12\left[\frac{0.55\,\mu}{\lambda}\right]^{1.3} e^{-H/1.5}\left[1 - \frac{0.32}{\ln(S)}\right]^{4/3}[1 + 0.33 sin(\alpha_S)] \ ,$$

$$\sigma_a = 0.01 + 0.4\, k_a \qquad \textbf{(6 a,b)}$$

Here, S is the relative humidity and σ_a is the uncertainty in k_a. There are additional trends with latitude and date, while the humidity term depends on aerosol type. The numerical variation around

these estimates is skewed by the likelihood of having much better haziness being smaller than the likelihood of having much worse haziness.

There may be additional sources of extinction not included in the above discussion, such as volcanic aerosol layers (see Meinel and Meinel 1983, Lockwood and Thompson 1986, and Rufener 1986 for typical effects) and urban pollution (see Joseph and Manes 1971, Husar and Holloway 1984, and Flowers, McCormick, and Kurfis 1969 for typical values).

Table 4. Extinction Coefficient

Site	Elevation (meters)	<k> winter	<k> summer
Mauna Kea, HI	4205	0.11	0.11
Los Angeles, CA	100	0.28	0.46
Kitt Peak, AZ	2064	0.15	0.21
Tucson, AZ	770	0.22	0.28
Tallahasee, FL	40	0.24	0.44
Atlantic City, NJ	10	0.27	0.47
Boston, MA	150	0.23	0.31
Athens, Greece	107	0.25	0.31
Jerusalem	775	0.18	0.28
South Pole	3000	0.14	0.14

The use of the equations above will allow for the extinction to be reasonably estimated for any site in the world. Nevertheless, the extinction varies unpredictably from site-to-site. Therefore, for specific applications, it would be best to have actual data from or near the site of interest. Appropriate data is available in the literature for much of the world. I have been able to find reliable extinction measures spread throughout the year for 225 locations scattered over all of the continents. Thus, any land site is usually within several hundred miles of someplace that has provided real data to the extinction problem. Publications with reliable and useful measurements from multiple locales include Abbott (1908-1954), Charlson *et al.* (1974), Joseph and Manes (1971), and Yamamoto, Tanaka, and Arao (1968). In Table 4 are presented average extinctions from a variety of sites. The total extinction coefficient will be the sum of the various contributing components,

$$k = k_R + k_{oz} + k_a \quad , \tag{7}$$

which represents the dimming towards the zenith. In other directions, the total loss in brightness will be given by:

$$\Delta m = k X \quad ,$$
$$\Delta m = k_R X_e(8.2\ km) + k_{oz} X_L(20\ km) + k_a X_e(1.5\ km) \quad , \tag{8 a,b}$$

in units of magnitudes. Equation 8a is valid for positions away from the horizon, while Equation 8b is valid anywhere, yet is required near the horizon for accurate answers. The apparent intensity of a source viewed through the atmosphere is given by the expression

$$I = I^* 10^{-\Delta m/2.5} \quad , \tag{9}$$

where I^* is the brightness above the atmosphere and I is the apparent brightness for the observer.

For an object low on the horizon, the apparent brightness varies roughly as $10^{-kX/2.5}$ with $X \approx 40$, so that even slight variations or uncertainties in k will change the apparent brightness by a large factor. For example, I will change by 45% as k changes from 0.24 to 0.25 magnitudes per air mass. The dominant source of uncertainty in k is from the aerosol component, and σ_a is typically larger than 0.05 magnitudes per air mass or higher, so that the deduced value of I will vary by roughly an order of magnitude or more. Thus, the uncertainty in the aerosol extinction is usually the dominant error in near-horizon visibility.

2.5 Source brightness

A fundamental fact for visibility of any astronomical object is its apparent brightness above the atmosphere. The brightness can be expressed as a magnitude, for which the visual magnitude (m_v) is relevant for normal day vision. This visual magnitude is related to the illuminance outside the atmosphere (I^*) as measured in foot-candles by

$$m_v = -16.57 - 2.5 \log (I^*) \quad . \tag{10}$$

The following equivalencies for "foot-candles" may be derived from Allen (1976).

$$\begin{aligned} 1\, \textit{foot-candle} &= 10.76\ \textit{lux} \\ &= 10.76 \times 10^{-4}\ \textit{phot} \\ &= 1\ \textit{lumen ft}^{-2} \\ &= 4.240 \times 10^{6}\ \textit{stars of } m_v = 0.00 \quad . \end{aligned}$$

If the source is extended, then its illuminance can be obtained by integrating its luminance, B (the surface brightness in units of nanoLamberts) over the source, *i.e.*:

$$I^* = 2.95 \times 10^{-7} \int B\, d\Omega \tag{11}$$

where Ω is the solid angle in steradians (Holladay 1926).

The following equivalencies for "nanoLambert" may be derived from Allen (1976) and Garstang 1989).

$$\begin{aligned} 1\ \textit{nanoLambert} &= 3.18 \times 10^{-10}\ \textit{stilb} \\ &= 3.18 \times 10^{-10}\ \textit{candela cm}^{-2} \\ &= 3.18 \times 10^{-6}\ \textit{nit} \\ &= 10^{-5}\ \textit{apostilb} \\ &= 4.61 \times 10^{-4}\ \textit{stars of } m_v = 0.00 \\ &= 26.33\ \textit{magnitudes per square arc-second}. \end{aligned}$$

The brightness and color of stars and planets have been tabulated in many books, with the *Astronomical Almanac* or *Astrophysical Quantities* (Allen 1976) as reliable references. The brightness of a wide range of sources is presented in Table 5, overleaf. The Sun's brightness, spectral energy, and sunspot information are well presented in Allen (1976). The Moon's m_v is given by

$$m_{moon}(\alpha) = -12.73 + 0.026\,|\alpha| + 4 \times 10^{-9}\,\alpha^4 \quad , \tag{12}$$

where α is the phase angle of the Moon in degrees, with $\alpha = 0°$ for full Moon (Allen 1976). The spectral reflectivities for the Moon are given in McCord and Johnson (1970). The lunar surface brightness is given by the highly complex Hapke equations (Hapke 1984) and the parameters given by Helfenstein and Veverka (1987). The luminance of the Moon's dark side, B_{ES} (*i.e.*, of the Earthshine) is given in nanoLamberts by:

$$B_{ES} = 2.59 \times dex\,[-0.4\,m_{moon}(\alpha')] \quad . \tag{13}$$

where $\alpha' = 180° - \alpha$ (Schaefer, Bulder, and Bourgeois 1992). Equation 13 was derived with an average lunar albedo of 0.067, whereas albedos for particular features can be found in Minnaert (1961).

Table 5. Source Brightness

Source	m_v	f (foot-candles)
Sun	-26.74	1.17×10^{4}
Setting Sun	-16	5.9×10^{-1}
Full Moon	-12.73	2.91×10^{-2}
Quarter Moon	-10.13	2.65×10^{-3}
Supernova (1000 pc)	-8	3.7×10^{-4}
Crescent Moon (a=135°)	-7.89	3.37×10^{-4}
Venus (at brightest)	-4.6	1.6×10^{-5}
60W bulb (860 lumens, 1 km)	-3.6	6.4×10^{-6}
Sirius	-1.46	9.04×10^{-7}
Vega	0	2.36×10^{-7}
Polaris	2.02	3.66×10^{-8}
Day vision limit	4.1	5.4×10^{-9}
Typical naked eye limit	6	9.4×10^{-10}
Typical binocular limit	9	5.9×10^{-11}
Brightest quasar (3C 273)	13.0	1.5×10^{-12}
Pluto (at opposition)	13.7	7.8×10^{-13}
Approximate HST limit	26	9.4×10^{-18}

2.6 Sky brightness

The brightness of the sky varies over a range of seven orders of magnitude. These variations are complex functions of the zenith distances of the Sun, Moon, and sky direction (Z_{sun}, Z_{moon}, and Z), the separation between the sky direction and the Sun and Moon (ρ_{sun} and ρ_{moon}), the altitude of the Sun (h_{sun}), the apparent magnitude of the Moon (m_{moon}), and the extinction coefficient (k). Ideally, the sky brightness should be calculated with multiple scattering, a round

Earth, and realistic scattering as a function of height in the atmosphere. Unfortunately, such studies (see Rozenberg 1966, Pyaskovskaya-Fesenkova 1957) only provide results of such complexity as to have small practical utility. Fortunately, with some simplifications it is possible to estimate the brightness of the cloudless sky at any time to perhaps 20% accuracy. The scattering of light by the atmosphere is the primary source of skylight; an approximate phase function describing this phenomenon is:

$$f(\rho) = 10^{5.36}[1.06 + \cos^2(\rho)] + 10^{6.15-(\rho/40^\circ)} + 6.2 \times 10^7 \rho^{-2} \qquad \textbf{(14)}$$

with ρ measured in degrees (Krisciunas and Schaefer 1991). The sky brightness in *nanoLamberts* (*nL*) is given by the following expressions:

$$B_{night} = B_o\left[0.4 + \frac{0.6}{\sqrt{1 - 0.96\sin^2 Z}}\right]10^{-0.4kX(Z)}$$

$$B_{moon} = f(\rho_{moon})\,dex\{-0.4[m_{moon} + 16.57 + kX(Z_{moon})]\}(1 - 10^{-0.4kX(Z)})$$

$$B_{twi} = \max\{1, dex[(\rho_{Sun}/90^\circ) - 1.1]\}\,10^{8.45 + 0.4h_{Sun}}(1 - 10^{-0.4kX(Z)})$$

$$B_{day} = 11{,}700 f(\rho_{Sun})\,dex[-0.4kX(Z_{Sun}))](1 - 10^{-0.4kX(Z)})$$

$$B_{sky} = B_{night} + B_{moon} + min[B_{twi}, B_{day}] + B_{city} \quad .$$

(15 a,b,c,d,e)

B_o is a parameter which varies with the time of day, the solar cycle, volcanic eruptions, and other phenomena (Krisciunas 1990); a typical value for a dark site B_o is 180 nL (Schaefer 1990b). The evaluation of B_{city} from first principles is a complex problem, with Garstang (1989) giving a definitive solution.

For some problems, an empirical approach might be preferable to the above theoretical calculation, so it is useful to have a series of measures under many conditions for comparison. The brightness of the daytime sky is given by Weaver (1947), Koomen *et al.* (1952), and Tousey and Hulburt (1948). The night sky brightness is given observationally by

Table 6. Sky Brightness Values

Source	B_{sky} (nL)
Sun's surface	6×10^{14}
Full Moon's surface	1×10^9
Typical daytime sky	5×10^8
Quarter Moon's surface	3×10^8
Zenith at sunset	3×10^7
Typical sky in big city	1×10^6
Zenith at Civil Twilight	1×10^5
Day vision/night vision	1500
Typical sky at Full Moon	1400
Zenith at Nautical Twilight	300
Near horizon for dark sky	240
Zenith for dark sky	20
Darkest ever observed	54

Krisciunas *et al.* (1987), Pilachowski *et al.* (1989), and Garstang (1989), while Schaefer (1990b) gives general formulas and methods for estimation. The sky brightness during twilight has been observed by Koomen *et al.* (1952). The effects of city light pollution have been observationally addressed by Walker (1973), Garstang (1989) and Lockwood, Floyd, and Thompson (1990). Sky brightness values under typical conditions are summarized in Table 6.

2.7 Glare

Glare occurs if a bright source of light masks a nearby fainter source. The paper by Holladay (1926) remains the best and most comprehensive report on the physiological measurement of the effects of glare. His basic conclusion was that glare can be treated merely as an increase in background light. This arises when light from the glare source is scattered (in the atmosphere, the telescope, or the eyeball) so that the background light around the source of interest rises significantly. The scattering in the telescope can arise from both diffraction and scattering on the mirror. Using the following expressions, Schaefer (1991c) calculated the added background from the four types of scattering: atmospheric (*atm*), diffractive (*dif*), mirror (*mir*), and eyeball (*eye*).

$$B_{atm} = 6.25\times10^{7} I^{*} \theta^{-2} [10^{-0.4kX} - 10^{-0.8kX}] \ ,$$
$$B_{dif} = 1.13\times10^{6} I^{*} D^{-1} \theta^{-3} 10^{-0.4kX} \ ,$$
$$B_{mir} = 2.60\times10^{8} I^{*} e^{-[\theta/0.4]^2} \ ,$$
$$B_{eye} = 4.63\times10^{7} I^{*} \theta^{-2} 10^{-0.4kX} \quad \textit{and}$$
$$B_{glare} = B_{atm} + B_{dif} + B_{mir} + B_{eye} \ .$$

(16 a,b,c,d,e)

Here, θ is the angle to the glare source (in units of *degrees*) and D is the aperture (in units of *cm*). If θ is less than 0.002° (≈ 6"), then these equations must be modified as described by Schaefer (1991c). If the glare source is extended, then additional modifications as prescribed by Schaefer, Bulder, and Bourgeois (1992) are needed. For viewing without a telescope, the values of both B_{dif} and B_{mir} should be set to zero.

The value of background brightness, B_{back}, from the sky and glare is, then, given by:

$$B_{back} = B_{sky} + B_{glare} \ .$$

(17)

Other additive terms (*e.g.*, from nebulosity) might have to be included in Equation 17.

2.8 Shadows

Shadows consist of an umbral region where the light source is completely hidden and a penumbral region where the light source is only partially hidden. The light source is blocked by an occulter and the shadow is cast upon a screen. For the Sun, the width, W, of the penumbral region on the screen can be expressed as:

$$W = \frac{2 L \alpha_{sun}}{\cos(\eta)} \quad , \tag{18}$$

where L is the distance between the occulter and the screen, α_{sun} is the angular radius of the Sun in radians (0.00465 radians), and η is the angle between the screen's normal and the Sun. For a straight edged occulter, the illuminance of the screen in the penumbra, I_{pen}, is given by:

$$I_{pen} = \left(\frac{I_{sun}}{\pi}\right)[\chi - \cos(\chi)\sin(\chi)]$$

$$\cos(\chi) = 1 - \left(\frac{2d}{W}\right) \quad , \tag{19 a,b}$$

where d is the perpendicular distance to the umbra/penumbra line, and χ is in radians. For $d < 0$ (in the umbra) the illuminance is *zero*, while for $d > W$ (in direct sunlight) the illuminance equals the direct solar illuminance, I_{sun}, the value for which is:

$$I_{sun} = 11700 \times 10^{-0.4kx} \tag{20}$$

(in units of *foot-candles*) as found from Equation 9 and Table 5. In the special case of a pinhole camera, the illuminance value of the image, $I_{pinhole}$, is given by:

$$I_{pinhole} = I_{sun}\left(\frac{D}{2 L \alpha_{sun}}\right)^2 \quad , \tag{21}$$

where D is the symbol for the aperture diameter of the pinhole and L stands for the distance between the pinhole and the screen.

A screen will also be illuminated by the skylight or scattered light off the surrounding terrain. Ideally, this extra light can be calculated with Equation 11 integrating the sky brightness from Equation 15, but this is difficult to do accurately. Nevertheless, an approximate answer is adequate for most applications. Koomen *et al.* (1952) present the illuminance on a flat surface pointed in various directions and times. For a horizontal surface, the illuminance value (in foot-candles) will be 3000 in daytime shadows, 40 at sunset, and 0.125 at civil twilight.

The screen will scatter some fraction of the incident light in all directions. For a perfectly diffusing (Lambertian) surface, the surface brightness of a screen, B_{screen}, illuminated with a total intensity of I will be given by:

$$B_{screen} = 1.076 \times 10^6 A I \quad , \tag{22}$$

where A is the screen's albedo, I is in *foot-candles*, and B_{screen} is in *nanoLamberts*.

2.9 Resolution

A distant point source of light can be broadened by atmospheric turbulence, diffraction by optics, and smearing over the aperture of a pinhole camera. Ideally, these effects should be superposed by the calculation of a two-dimensional convolution of the original image with all the smearing functions. In practice, it is difficult to calculate the convolution and impossible to present the result in a simple and general manner. As an alternative, it is possible to calculate the characteristic area of the convolution, which can be expressed as the second moment (μ) of the apparent brightness distribution. It can be proved that the second moment of a convolution of circularly symmetric normalized distributions is just the sum of the second moments of the functions being convolved. Thus (Schaefer 1993b), the second moment of the smearing function, μ_{smear}, is determined by the following expressions:

$$\mu_{seeing} = 0.361\,(M\alpha_{seeing})^2 \quad ,$$

$$\mu_{diff} = 2\left(\frac{\lambda M}{D}\right)^2 = \left(\frac{16.0'' cm\, M}{D}\right)^2 \quad ,$$

$$\mu_{aper} = \frac{1}{8}\left(\frac{M D}{L}\right)^2 = \left(\frac{72900'' M D}{L}\right)^2 \quad ,$$

$$\mu_{smear} = \mu_{seeing} + \mu_{diff} + \mu_{aper} \quad .$$

(23 a,b,c,d)

Here, M is the magnification of any optics being used, α_{seeing} is the FWHM [1] of the seeing disk, λ is the wavelength of light (with $\lambda = 0.55$ microns for the right hand side of Equation 23b), D is the aperture of the optics, and L is the length of projection, both in the same units. For viewing with no optical aid, the values of both μ_{diff} and μ_{aper} are zero; alternatively, for viewing with focusing optics, the μ_{aper} value is zero. The distributions in Equation 23 are normalized, which implies that no photons are lost in the smearing processes. The second moment of the smeared image will be given by:

$$\mu = \mu_{smear} + \mu_{source} \quad .$$

(24)

When calculating the second moment of the original source (μ_{source}), the distribution must be normalized. For poorly resolved images or images that are fairly uniform, the source visibility can be equated to that of a uniform circular disk of identical second moment with a diameter

$$\zeta = \sqrt{8\,\mu}$$

(25)

[1] *FWHM* = Full Width at Half-Maximum of the distribution function.

This disk of equal visibility must be normalized in height by the same factor as was used to normalize the original source distribution.

For pinhole cameras, there will be some pinhole diameter which will optimize the resolution. This optimal diameter can be found by minimizing the quantity ($\mu_{diff} + \mu_{aper}$) with respect to D. Thus, the resolution is best when $D^2 = 4\lambda L$, which corresponds to the pinhole occupying exactly one Fresnel zone (Young 1972).

The seeing disk towards the zenith has a typical FWHM of 1.5" during the night or 3" during the day. Near the horizon, the seeing disk is larger because the light traverses a greater path length of turbulent air. For a sightline 60° from the zenith, which passes through twice as much air as the zenithal path, the second moment of the seeing disk will double. To generalize:

$$\alpha_{seeing} = \alpha_S^* \sqrt{X} \quad , \tag{26}$$

where α_S^* is the FWHM of the seeing disk at the zenith and X is the air mass along the line of sight (from Equation 3a for $H_e = 8.2$ km).

The human eye has an intrinsic resolution or pixel size (θ_{CVA}) that depends on the light level. The "critical visual angle" or pixel diameter is given by:

$$\theta_{CVA} = \frac{40''}{S} dex\left(8.28\, B^{-0.29}\right) \qquad if\ log(B) > 3.17\,,$$

$$=$$

$$\theta_{CVA} = min\left[900'', \frac{380''}{S} dex\left(0.3\, B^{-0.29}\right)\right] \qquad if\ log(B) < 3.17\ . \tag{27 a,b}$$

(*c.f.* Schaefer, Bulder, and Bourgeois 1992). Here, B is the brightness (in units of *nanoLamberts*) of the source of interest, and S is the *Snellen ratio* (the *Snellen ratio* is unity for 20/20 vision, while $S = 2.0$ for good 20/10 vision) of the observer.

How far apart must two identical sources be for the eye to resolve them? The traditional answer is *Rayleigh's criterion*, where sources are presumed resolved if

$$\theta > \frac{1.22\,\lambda}{D} \tag{28 a}$$

where θ is the angular separation of the centers in radians, λ is the effective wavelength of the light beam, and D is the aperture of the telescope. In the amateur astronomy community, the popular *Dawes' Limit* gives a similar empirical result:

$$\theta > \frac{4.5}{D} \tag{28 b}$$

where D is measured in inches and θ in arc-seconds. Even though these criteria are precisely stated, the problem is poorly defined. In particular, a wider separation is needed if background must be seen between the two sources to prove duplicity (as opposed to elongation), while deviations from a point source shape can be detected with much smaller separations for bright sources. Another problem with these limits is that they only apply when the image is bright and diffraction dominates. Two examples where Dawes' limit does not yield correct results are that most naked eye observers ($D \approx 0.2$ inch) cannot resolve the major separation of the bright binary stars ϵ^1 and ϵ^2 *Lyræ* despite their 208" angular distance from each other and that an observer with a 10-inch telescope aperture and poor seeing (say, $\alpha_{seeing} = 5''$) cannot resolve either ϵ^1 or ϵ^2 *Lyræ* with their 2.6" angular separations. Perhaps the best rule is that two similar images can be "resolved" if their separations are comparable to the characteristic size of the perceived image.

2.10 Optics

For questions of visibility in modern times, a model of telescope optics may be needed. The optics of telescopes have well known effects, however, the researcher must beware the many subtle phenomena that can change the result. The best discussions from a visibility point of view are in Tousey and Hulburt (1948) and Schaefer (1990b). The optics of a telescope will affect both the perceived source intensity as well as the perceived background brightness. These effects can be modelled as (Schaefer 1990b, but see minor corrections in Schaefer, Bulder, and Bourgeois 1992 and as below)

$$I^* = I F_I \quad ,$$
$$B = B_{back} F_B \quad .$$

(29 a,b)

The forms for the optics correction factors, F_I and F_B, are the result of many corrections for individual effects with

$$F_I = F_b F_t F_p F_a F_{SC} F_c F_e F_r \quad ,$$
$$F_B = F_b F_t F_p F_a F_{SC} F_c F_m \quad .$$

(30 a,b)

The individual components of the correction factors are evaluated with use of the following expressions:

$$F_b = 1 \qquad \textit{if binocular vision},$$
$$F_b = \sqrt{2} \approx 1.41 \qquad \textit{if monocular vision};$$

(31 a,b)

$$F_t = \frac{1}{T_{tel}} \quad ,$$

(31 c)

$$F_p = max\left(1, \left[\frac{P_{st}}{M P}\right]^2\right) \quad ,$$

(31 d)

$$F_a = \left(\frac{P_{at}}{D} \right)^2 \qquad \textbf{(31 e)}$$

$$F_r = \left(1 + 0.03 \left[\frac{M\zeta}{\theta_{CVA}} \right]^2 \right) S^{-2} \quad , \qquad \textbf{(31 f)}$$

$$F_m = M^2 \quad ; \qquad \textbf{(31 g)}$$

the expressions for the factors F_c, F_e, and F_{SC} depend upon the flux-level; the following constraints apply:

Case #1 - *log(B) > 3.17*

$$F_e = 10^{0.4kX} \quad ,$$

$$F_{SC} = min\left[1, \left(\frac{D}{P_{at} M} \right)^2 \frac{\left(1 - e^{-(P_{at}/0.62\,cm)^2} \right)}{\left(1 - e^{-[D/(0.62\,M\,cm)]^2} \right)} \right] \quad ,$$

$$F_c = 1 \quad ; \qquad \textbf{(31 h,i,j)}$$

Case #2 - *log(B) < 3.17*

$$F_e = 10^{0.48kX} \quad ,$$

$$F_{SC} = min\left[1, \frac{\left(1 - [P_{at}/12.44]^4 \right)}{\left(1 - [D/(12.44\,M)]^4 \right)} \right] \quad ,$$

$$F_c = 10^{-0.4(1-[B-V]/2)} \quad . \qquad \textbf{(31 k,l,m)}$$

The correction factors' subscripts are: *b* for binocular vision, *e* for atmospheric extinction, *t* for transmission through the telescope, *p* for light lost outside the pupil, *a* for light gathering power of the telescope aperture, *r* for the resolving power of the eye, *SC* for Stiles-Crawford effects, *c* for source color, and *m* for the magnification. Here, T_{tel} represents the transmission of the telescope (a typical value for which is 0.8, according to Schaefer 1990b), *S* stands for the Snellen ratio, *M* refers to the magnification of the optics, *D* to the aperture of the optics, and [*B-V*] denotes the visual color-index of the source; the numerical factor "12.44" has units of *centimeters*. The diameter of the observer's pupil is given by:

$$P = max\left\{0.2\,cm, (0.7\,cm)\,exp\left[-0.5\left(\frac{Y}{100}\right)^2 - 0.16\left(B\times 10^{-6} + \frac{0.172\,I}{1.122^{M\theta}}\right)^{0.4}\right]\right\} \quad . \tag{32}$$

Here, Y represents the observer's age in years, B the background brightness in *nanoLamberts*, and I the intensity of a glare source placed an angle θ in *degrees* away with a magnification of M. For a standard observer ($Y = 25$ *years*) with no glare source,

$$P_{st} = \max\{0.2\,cm, (0.68\,cm)\exp[-0.16\,(B\times 10^{-6})^{0.4}]\} \quad , \tag{33}$$

which is an appropriate expression for comparison with the observers contributing to the physiological data on visibility thresholds.

2.11 Thresholds of Human Vision

All of the preceding tools can be used to calculate the apparent brightness of the astronomical source as well as the surface brightness of the background. The question of visibility then depends on the physiological question of can such a source be detected by the human eye. Extensive experimentation under a wide variety of conditions was made in the 1940's and 1950's, and it is this work which can be used to quantify visibility thresholds.

Operationally, the results can be divided into the detection of point sources and uniform extended sources. For the detection of point sources of light, Hecht (1947) gives a convenient and accurate formula for the threshold illuminance value, I_{th}, namely:

$$I_{th} = c\,(1 + \sqrt{kB}\,)^2 \quad , \textit{ where}$$
$$\log(c) = -8.35 \textit{ and } \log(k) = -5.90, \textit{ if } \log(B) > 3.17, \textit{ or}$$
$$\log(c) = -9.80 \textit{ and } \log(k) = -1.90, \textit{ if } \log(B) < 3.17. \tag{34 a,b,c}$$

Here, B represents the effective background in nanoLamberts, as taken from Equation 29b. For uniform extended sources, the source visibility is characterized with a contrast ratio, C, defined as

$$C = \frac{|B_{source} - B|}{B} \quad , \tag{35}$$

where B_{source} represents the surface brightness of the source to be detected. Note that C does not depend on whether the source is brighter or darker than the background. The comprehensive and vast study by Blackwell (1946) remains the definitive work. He reports on the thresholds for highly-experienced young observers of average vision allowed to use binocular vision with natural pupils and allowed to choose averted vision or direct fixation and allowed a leisurely examination.

For uniform sharp-edged circular regions of angular diameter ζ, the threshold contrast ratio, C_{th}, is given by:

$$C_{th} = \max\left[2.4\,B^{-0.1}, 20\,B^{-0.4}\right]\left(\frac{\theta_{CVA}}{\zeta}\right)^2 \quad \text{if } \theta_{CVA} > \sim\zeta \text{ , or}$$

$$C_{th} = 0.0028 + 2.4\,B^{-0.1}\left(\frac{\theta_{CVA}}{\zeta}\right)^2 \quad \text{if } \log(B) > 6 \text{ , or}$$

$$\log(C_{th}) = -[0.12 + 0.40\log(B)] + [0.90 - 0.15\log(B)]\log(100'/\zeta)$$
$$\text{if } \log(B) < 6 \text{ and } \zeta > \sim 20' \quad .$$

(36 a,b,c)

These three formulas are empirical fits to the data in Blackwell, and they reproduce his values to 5%. Equation 36a is an expression of "Ricco's Law," in that the threshold is proportional to ζ^{-2}. Corrections to these values are needed for an observer's experience, as discussed in Schaefer (1993b, 1990b).

Near the sensitivity threshold of the human eye, detection is a probabilistic question; see Lamar *et al.* (1947, 1948) for a definitive analysis. The quoted physiological thresholds are always for a 50% probability of detection.

2.12 Color Vision

The human eye can detect colors from sources brighter than 1500 nL. The reason is that *photopic vision* (*i.e.*, day vision which uses the retinal cones; as opposed to *scotopic vision*, or night vision, which uses rods) has three types of photopigments, each with a different spectral sensitivity. So the eye simultaneously measures the brightnesses of an image over three different wavebands, much like taking CCD images through three different filters or taking a color photograph with three different chemical dyes. The situation is also analogous to photometry in the Johnson *UBV* system, where three intensities are measured (or alternatively, one brightness and two colors). For each photopigment, the effective brightness will be

$$X = \Lambda \int x^* E_\lambda \, d\lambda \quad ,$$

$$Y = \Lambda \int y^* E_\lambda \, d\lambda \quad ,$$

$$Z = \Lambda \int z^* E_\lambda \, d\lambda \quad .$$

(37 a,b,c)

The numerical value of the constant Λ is *6.80 lumen s erg*$^{-1}$. The values of x^*, y^*, and z^* are given as a function of wavelength, λ, in Table 7 (for example, see Cornsweet 1970 or Weast 1974, for

tables with higher spectral resolution). The brightness E_λ has units $erg/cm^2/s/\AA$. With these units, Y will have cgs units of *phot*, so that

$$m_v = -23.99 - 2.5 \log(Y) \quad . \tag{38}$$

Table 7. Colorimetric Coefficients

λ (Å)	x*	y*	z*
3750	0.0006	0.0000	0.0020
4000	0.0143	0.0004	0.0679
4250	0.2148	0.0073	1.0391
4500	0.3362	0.0380	1.7721
4750	0.1421	0.1126	1.0419
5000	0.0049	0.3230	0.2720
5250	0.1096	0.7932	0.0573
5500	0.4334	0.9950	0.0087
5750	0.8425	0.9154	0.0018
6000	1.0622	0.6310	0.0008
6250	0.7514	0.3210	0.0001
6500	0.2835	0.1070	0.0000
6750	0.0636	0.0232	0.0000
7000	0.0114	0.0041	0.0000
7250	0.0020	0.0007	0.0000
7500	0.0003	0.0001	0.0000

The intensities in Equation 37 are usually normalized as

$$x = \frac{X}{X + Y + Z} \quad ,$$

$$y = \frac{Y}{X + Y + Z} \quad ,$$

$$z = \frac{Z}{X + Y + Z} \quad . \tag{39 a,b,c}$$

The two values x and y are then a complete description of any color (the value of z is ignored as being redundant). The color white (and the color of the Sun) is characterized as $x = 0.33$ and $y = 0.33$. The color can be parameterized with

$$\Psi = ATAN2(y - 0.33, x - 0.33) \quad , \tag{40}$$

where the *ATAN2* function is the arctangent computer-subroutine which picks the correct quadrant.

The definition of the primary colors is always quoted differently in "standard" references, but a set of reasonable choices is given in the following tabulation (Table 8). Just the same color vision in humans is quite complex with many nonlinear effects, and the above must serve only as a primer. For good detailed descriptions, I recommend Cornsweet (1970).

2.13 Weather statistics

For any phenomenon involving time or rates of time, the frequency of clear skies may be important. In general, relevant data can generally be obtained from local airports, newspapers, or a national meteorological center. Schaefer (1987a) reports on cloudiness statistics low on the twilight sky near sunrise and sunset; he found that good and average sites have cloudiness probabilities of a third and of a half, respectively. The clarity odds from one night to the

next are correlated, since it usually takes a few days for a bad weather system to pass by. Thus, a cloudy night will be followed by a cloudy night 75% of the time (for both good and bad sites), while a good site will have a cloudy night follow a clear night 25% of the time (50% for an average site). In any study of old historic events, it should be remembered that local weather patterns might have changed significantly in the ensuing years.

Table 8. The Primary Colors: their λ's and Ψ's.

Color	red	orange	yellow	green	blue	violet
λ Range (Å)	> 6100	5900-6100	5700-5900	5000-5700	4400-5000	< 4400
Ψ Range	< 0°	0° - 21°	21° - 63°	63° - 148°	148° - 242°	> 242°

3. Results

The previous section gives explicit formulas which describe the phenomena affecting visibility. To be useful, however, these equations must be put together in a manner that makes physical sense. For example, to calculate the heliacal rise date of Sirius from ancient Egypt, the parameters needed are the position of Sirius, the brightness of the twilight sky, the likely extinction from Egypt, the air mass, and the thresholds of vision. In this case, the repeated visibility testing for Sirius throughout the dawn twilight and for many days introduces a complexity best handled with a computer program. Other results may be presented best as simple formulas or as a series of equations. In this section are presented results on a variety of celestial visibility problems, with each result having applications to many historical and astronomical questions. In two cases (for sunspot visibility and the extinction angle), I will give detailed expositions on how the "tools" can be used to create working models.

3.1 Sunspots

Large sunspots are easily visible to the human eye (with proper protection) and have been extensively recorded by the ancient Chinese (Yau and Stephenson 1988). Until recently, the best rule of thumb for sunspot visibility has been that a sunspot must be larger than one arc-minute in size to be visible (Eddy 1980, Eddy, Stephenson, and Yau 1989). This rule has been strongly refuted by observation (Mossman 1989, Keller 1980, Keller 1986, and Schaefer 1991d).

Schaefer (1991d and 1993b) has developed a detailed model that predicts the visibility of sunspots with direct vision, with telescopic assistance, with a pinhole camera, and with telescopic projection. Here, I will give the derivation of the direct vision limit in detail.

First, estimate the surface brightness of the Sun after it has been dimmed enough for safe viewing. An approximate answer is adequate, since the dependence on B in Equation 36b is so weak.

Frequently, the setting Sun appears dim enough for good visibility, and this dimming can be estimated roughly with $X = 40$ (for near the horizon from Equation 2b) and $k = 0.3$ (a typical value, see Table 4) so the Sun's surface brightness will dim by a factor of $10^{-4.8}$ (from Equations 8a and 9). For a more accurate calculation, Equation 8b should be used with, say, $k_R = 0.10$, $k_{oz} = 0.03$, and $k_a = 0.17$, and air masses for $Z = 90°$ from Table 3, so that the dimming factor is $10^{-7.1}$. Alternatively, the recommended filter for safe solar observing is welders' glass #14 (Chou 1981ab) which has a transmission of e^{-14} or $10^{-6.1}$. Thus, the surface brightness of the Sun will be reduced from 6×10^{14} nL (see Table 6) to roughly 10^{8} nL.

Second, the second moment of the smearing must be calculated. For direct vision, $\mu_{diff} = 0$, $\mu_{aper} = 0$, and $M = 1$; so, from Equation 23:

$$\mu_{smear} = 0.361\ \alpha^2_{seeing} \quad . \tag{41}$$

A typical daytime value for α_{seeing} is $3''$, so that μ_{smear} is $(1.80')^2$.

Third, the second moment of the sunspot structure must be calculated. Following Allen (1976), I will take the fraction of the sunspot area occupied by the umbra to be 17% on average. I will also idealize the umbra and penumbra as concentric circles with contrast values of 0.91 and 0.26, respectively. (Schaefer [1993b] demonstrates that the visibility is only weakly dependent on these assumptions.) Thus the contrast value is 0.91 for $\theta < 0.41 R_{spot}$, 0.26 for $0.41 R_{spot} < \theta < R_{spot}$, and zero for $\theta > R_{spot}$, where θ is the angular distance from the spot's center. But these contrasts must be normalized by a factor N so that the second moments can be added as in Equation 24. Thus

$$1 = 2\pi \int [N C_{spot}(\theta)]\, \theta\, d\theta = 1.16\, N R^2_{spot} \quad . \tag{42}$$

The value of the second moment of the contrast, which characterizes the solid angle covered by the sunspot, is then given by

$$\mu_{source} = 2\pi \int \theta^2 [N C_{spot}(\theta)]\, \theta\, d\theta = 0.44\, N R^4_{spot} = 0.38\, R^2_{spot} \quad . \tag{43}$$

Fourth, we need to find the properties of a uniform circular disk with the same visibility as the sunspot. The perceived size of the spot may be ca
lculated by:

$$\mu = 0.361\ \alpha^2_{seeing} + 0.38\, R^2_{spot} \quad , \tag{44a}$$

from Equations 24, 41, and 43. Given that $\alpha_{seeing} << R_{spot}$ for spots visible with unaided vision and for any reasonable seeing, we have, then:

$$\mu = 0.38\, R^2_{spot} \quad . \tag{44b}$$

The diameter of the uniform disk with equal visibility will have the same characteristic area as the sunspot, or from Equations 25 and 44b,

$$\zeta = 1.74\, R_{spot} \quad . \tag{45}$$

The total darkness in this disk must equal the total darkness in the (unnormalized) sunspot. Thus,

$$2\pi \int C_{spot}(\theta)\, \theta\, d\theta = 2\pi \int C_{disk}(\theta)\, \theta\, d\theta \quad . \tag{46a}$$

or

$$C_{disk} = 0.49 \quad . \tag{46b}$$

We now have enough information to check whether the uniform disk would be visible.

Fifth, the critical visual angle must be quantified. From Equation 27a, this is

$$\theta_{CVA} = \frac{44''}{S} \quad , \tag{47}$$

where S is value for the observer's Snellen ratio. For filters which change the Sun's surface brightness from 10^7 *nL* to 10^9 *nL*, the θ_{CVA} changes only from *42"/S* to *48"/S*. But the typical error in measuring S is 25%, so the uncertainties in not specifying the filter technique are relatively small.

Finally, the visibility of the uniform disk is evaluated with the data from Blackwell (1946). For an apparent surface brightness of 10^8 *nL* and the results of Equations 36b, 45, 46b, and 47, I find a threshold sunspot size of

$$R_{spot} = \frac{22''}{S} \quad . \tag{48}$$

Thus we see that sunspots smaller than one arc-minute in size may be detected with normal human vision. Schaefer (1993b) has collected well-determined observational limits from four observers and from one network of observers which validate the limit in Equation 48.

3.2 Stars and planets during the daytime

The stars and planets do not turn off when the Sun rises, it is just that the glare from the Sun is so overwhelming as to render all but the brightest of them invisible. How faint a source can be seen in the daytime? For a typical daytime sky brightness of 5×10^8 *nL* (Table 6) and extinction with $\Delta m = 0.3$ magnitudes, then Equations 9, 10, and 34 show the limit for normal vision to be *-3.1* magnitude. Thus under normal daytime conditions, only Venus can be seen without telescopic aid. If the Sun is close to the horizon (such that $B_{sky} = 3 \times 10^7$ *nL*, see Table 6), then the unaided normal eye can see down to *-0.3* magnitudes, which covers Venus, Jupiter, Mars, Sirius, and Canopus. Tousey and

Hulburt (1948) discuss the daytime visibility of stars with telescopes and from high altitude aircraft. Schaefer (1991c) analyzes the visibility of sungrazing comets and Venus near conjunction with and without telescopic assistance. For normal unaided vision,

$$m_{lim} = -9.28 + 5\log(\theta) \quad , \tag{49}$$

where θ is the angular distance from the Sun. Similar equations can be found for telescopic viewing.

There have been persistent myths that stars can be seen in daytime by looking up a tall chimney or by looking in the reflection of water in a deep well. The literature does contain such reports, but they are all literary devices, third-hand anecdotes, or childhood memories of old men (Hughes 1983, Perrachi 1993). However, all first-hand reports (Humboldt 1851, Hynek 1951, Smith 1955, Slabinski 1990, and Sanderson 1992) are emphatic on the impossibility of seeing stars under such conditions. In addition, strong theoretical arguments show that the glare reduction is insignificant, that the perpendicular reflectivity of water is 2%, and that the small background field-of-view actually makes the threshold up to a magnitude worse (Schaefer 1991c). Thus, both observation and theory strongly prove the chimney/well myths to be false.

3.3 Shadow bands

Shadow bands are ghostly and irregular dark lines that are visible in the moments before and after a total eclipse of the Sun (Marschall 1984). The bands arise from diffraction and scintillation of a long thin light source, with Codona (1986) providing the definitive theoretical work.

The phenomena of sunset shadow bands (Rozet 1906 and Ives 1945) and volcanic shadow bands (O'Meara 1986) are still mysteries.

3.4 Cardinal orientation

Many ancient monuments and modern cities are aligned with axes pointing in a north/south or east/west direction. Such a cardinal orientation must ultimately come from astronomical observations. In modern times, standard surveying techniques can establish north with arc-second accuracy. In older times, before good ephemerides and instruments were available, the methods (and motivations!) of cardinal alignment are generally not known. Many procedures have been proposed (Zaba 1953, Edwards 1961, Neugebauer 1980, Haack 1984, and Isler 1989), and these fall into three categories.

The first (and simplest) category is to sight along some line which points to the azimuth of a star that does not move. In modern times, Polaris is about a degree from the pole and it may be possible (with very-large-scale solidly-mounted sights and careful repeated observations) to interpolate between its extreme azimuths to perhaps a quarter of the polar distance. In ancient times, no stars were close to the pole so that the resultant accuracy would have been lower.

The second category involves methods for finding the azimuth of the shortest shadow. Around noon, the shadow length varies little so it is highly inaccurate to directly take the azimuth of the

shortest measured shadow. Instead, the azimuths of equal length shadows several hours before and after noon can be bisected to find north. For an accuracy better than roughly 0.2°, the observations must be made by direct observation of the Sun's edge (as opposed to trying to define some position within the shadow's penumbra) near the times of a solstice.

The third category involves the sighting of azimuths of rising and setting of some object and then bisecting the angle to find north. This last category has the severe problem that the altitude on both horizons must be identical (*cf.* Schaefer 1986). Another severe problem for high accuracy alignments by this method is that the natural variation of refraction will introduce an error in the measured azimuths (Schaefer and Liller 1990).

3.5 Sky colors

"Why is the sky blue?" is the classic young child's question that stumps their parents. A physicist would explain about Rayleigh scattering, where the molecules in our air preferentially scatter blue light. This is indeed the correct (although simplified) answer. Simple applications of Rayleigh scattering can explain most of the colors seen in the sky. For example, the red color of the setting sun is due to the scattering of all blue light out of the direct rays. The red colors of twilight are caused by the only direct illumination of the air being the low red Sun. The white color of the daytime sky close to the horizon is a result of the exact compensation of the skylight production by scattering with its extinction over a large optical path length. This can be looked at quantitatively where the luminosity of a volume at a distance x will be proportional to the optical depth from Rayleigh scattering (τ_R), while the extinction will be given by $exp(-x\tau_R)$. The apparent brightness will be an integral along the line of sight, which for a near horizon path is effectively $0\int^{\infty} exp(-x\tau_R)\, x\, dx$ and is a constant independent of wavelength.

The literature on sky colors is extensive. Chandrasekhar (1960) and Rozenberg (1966) present exhaustive mathematical analyses and bibliographies of scattering in the atmosphere. Hulburt (1953) gives a more accessible mathematical explanation for sky colors. Walker (1989) presents an excellent and thorough non-mathematical explanation for colors and brightnesses for the whole sky under many conditions. Minnaert (1954) and Meinel and Meinel (1983) give good phenomenological descriptions.

3.6 Rainbows and halos

As sunlight passes through the air, it is refracted, dispersed, and reflected by water drops and ice crystals. This scattered light appears from special directions so as to produce bright circles and spots on the sky. The most familiar phenomena are the "rainbow" seen after a passing storm (42° from the antisolar direction), the "halo" seen around the Sun or Moon when cirrus clouds are overhead (22° from the light source), the colored patches of light known as "sundogs" or "parhelia" (somewhat more than 22° from the Sun and at the same altitude), and the colorful circles cast onto clouds below an airplane called "glories" (extending typically several degrees around the antisolar direction). Many less-well-known effects exist, including "iridescent clouds," "coronas," "Bishop's rings," and

"circumzenithal arcs" which all give good displays of color. Scattering by ice crystals can produce a wealth of arcs, pillars, and halos, which for rare displays can light up the sky in an intricate spider's web of light. The frequency of rainbows and halos is surprisingly large, with observers from many locations logging some such phenomenon typically a third of the days throughout the year.

The literature on rainbows and halos is extensive. My favorite books are Minnaert (1954) for a general description of the many phenomena, Konnen (1985) for an impressive discussion of polarization effects, and Boyer (1959) for a detailed history and explanation of the rainbow. Despite the excellence of these sources, *Rainbows, Halos, and Glories* by Greenler (1980) is by far the best general reference. This book gives definitive and comprehensive explanations as told in a fun manner with many beautiful and unique pictures.

3.7 Mountain shadows

A sunrise viewed from the top of a tall mountain has an eerie beauty. Some mountains attract hundreds of tourists daily and have been turned into commercial ventures. However, few are the watchers who turn around after sunrise to see the shadow of the mountain, which appears as a conical dark region looming above the western horizon. As the Sun gets higher, the shadow cone flattens out and soon disappears. For an observer away from the top of the mountain, the structure can get quite complicated. The effects are most prominent from near the top of sharp, isolated, and tall peaks. The observer need not be a mountain climber to see the cone of darkness, since many suitable mountains have roads built to the top, including Mount Washington, NH; Pike's Peak, CO; Mount Evans, CO; Kitt Peak, AZ; Mauna Kea, HI; and Haleakala, HI.

The mountain's shadow is cast onto the haze in the air. For an observer at the peak, any line of sight below the antisolar direction will lie entirely in shadow so that the only light seen will be by scattering of the relatively faint skylight. Any line of sight above the antisolar direction will be entirely in sunlight, so that the scattered light from the bright Sun will result in a bright sky. From the top of a conical peak, the shadow will be perfectly sharp. Livingston and Lynch (1979) prove that the shadow from the top will always appear conical (regardless of the mountain's shape) due to perspective effects, although the shadow edge may get fuzzy; their paper (and Lynch 1980) gives a good mathematical model for the shadow brightness from anywhere on the mountain. A computer program for calculating shadow images from various locations near the top of a conical mountain is presented by Schaefer (1988a).

Mountain shadows are similar to the phenomena of "crepuscular rays," "sunbeams," and the "Earth shadow." These other shadow types are modelled in Lynch (1987) and Rozenberg (1966).

3.8 Green flash

The green flash appears when the last portion of the Sun's disk sinks below the horizon and changes to a beautiful emerald color. Green flashes can also be seen just before sunrise and can appear as an electric blue color if the air is especially clear. The phenomenon first came to the attention of the

scientific community in 1882 with the publication of a book by Jules Verne titled *Le Rayon Vert.* Good descriptions appear in Minnaert (1954) and Meinel and Meinel (1983), but the definitive work is O'Connell (1958).

The explanation for the green flash involves the normal dispersion and extinction of sunlight by the air. As sunlight enters the atmosphere, refraction bends the light path by around 0.5° around the time of sunset. Each color will suffer a different dispersion, with the blue light bent more than red light. The light for each color will appear as a disk, but due to dispersion, the red and yellow disks will be below the green and blue disks. As the Sun sets, the red and yellow disks will disappear first, leaving a brief period of time when only the green and blue light can be seen. Under normal conditions, the extinction is so heavy that no blue light can get through. Thus, for a short time, only green light is visible.

The basic mechanism was worked out early in the 1900's. However, it was not until Shaw (1973) and Menat (1980) that detailed calculations from first principles were able to reproduce the observations. Independently, I repeated their earlier work and embodied the relevant equations in a computer program that is applicable to any arbitrary planetary system and atmosphere (Schaefer 1992b). The causes of flashes during reddened sunsets and long duration flashes are not known.

3.9 Twilight

After sunset, a complex play of light and color unfolds in the sky. The general schedule of fading and shading is familiar to everyone, yet nevertheless, many surprising phenomena await any observant person. Around sunset, "crepuscular rays" may appear. These are bright and dark lines which converge on the Sun's position, much as in the flags of Arizona, Tibet, and the naval ensign of Japan. The patterns are caused by parallel bands of cloud or mountain shadows that converge on the Sun due to perspective effects. On particularly favorable evenings, these shadows can be traced through the meridian all the way to where they converge again opposite the Sun as "anti-crepuscular rays." Immediately after sunset, the "Earth shadow" begins to rise over the eastern horizon, and appears as a broad arc that defines the top of a dark blue/grey region where the line-of-sight looks entirely within the Earth's umbra. On occasion, the "Belt of Venus" appears just above the Earth shadow as a band of delicate pink, where the line-of-sight passes through regions of the atmosphere whose primary light source is the highly reddened setting Sun. The "purple light" appears as a diffuse lilac-tinted light roughly 30° above the setting Sun when the Sun itself is about 4° below the horizon. This subtle glow is caused by red sunset light scattered off a high aerosol layer being added to the normal blue scattered light to yield a purple color. "Arago's point" is the location some 25° above the anti-solar direction at which the polarization is zero. Above Arago's point, the polarization is parallel to the horizon as expected from Rayleigh scattering, but lower than this the polarization is perpendicular to the horizon probably due to multiple scattering effects. This point can easily be found before civil twilight ends by holding up two polarizing filters with different orientations and varying the altitude until the sky brightness is equalized.

Definitive descriptions of the many twilight effects are given in Minnaert (1954) and Meinel and Meinel (1983). The Meinels' book gives many beautiful color pictures, extensive discussion on the effects of volcanic eruptions, and a brilliant account of twilight phenomena seen on various bodies in our Solar System. Rozenberg (1966) presents an exhaustive treatment of the mathematical modelling of twilight colors and brightnesses.

3.10 Heliacal rise

The heliacal rising of a star or planet occurs on the first day that the object is visible in the dawn light after the conjunction with the sun. A model exists (Schaefer 1987a) for predicting the dates and azimuths of heliacal risings and settings based on the various mathematical tools discussed in this review. Estimates of the uncertainty in these quantities caused by normal variations of the moonlight and weather as well as uncertainties in the observing conditions are presented. In the same paper, 556 observations of heliacal rise are reported which confirm the theoretical calculations, with the few exceptions being directly caused by clouds near the horizon. The list-out of a short computer program, with the use of which the date of heliacal rising is calculated, has been published, also (Schaefer 1985b).

The achronal rising of a star is when the the star first appears in the eastern evening sky just after sunset. Unfortunately, the phenomenon is poorly defined. Did the ancient cultures mean that the star actually rose at sunset? If so, then the event is unobservable. Did the ancient cultures adopt the end of twilight as the time when the star must be visible? If so, then what was their definition of the end of twilight? The possibilities include civil, nautical, or astronomical twilight. These varying definitions can change the date of achronal rising by up to a month while the azimuth will vary by many degrees for temperate latitudes. As achronal risings are either invisible or highly uncertain in date and azimuth, their utility in archeoastronomical settings is problematic.

3.11 Young lunar crescents

Lunar calendars usually have the start of the month defined by when the thin crescent is first visible after new moon. The prediction of crescent visibility is a very difficult problem, for which astronomers since the Babylonian times have invested much effort (see Ilyas 1984). In modern times, the utility of a calendar is greatly reduced if dates cannot be predicted. (Try making a plane reservation for next month if the day-name for the date is unknown.) Each year, there are several conferences around the world that seek to address this problem. With more than 20% of the world population adhering to the Islamic faith, crescent visibility is the one astronomical problem that affects the greatest number of humans.

The systematization of lunar calendars is important in a world of widespread instantaneous communication. From an astronomical point of view, the Lunar Date Line concept (Ilyas 1984) has great merit. The Lunar Date Line is a roughly parabolic shaped curve drawn over the world on one side of which the Moon can be seen, while on the other side of which the lunar month must await one

more day to start. The question is then "What algorithm should be used to calculate the Lunar Date Line?"

Until recently, all prediction algorithms were based on observational data sets from one site (in modern times, all such algorithms are based on Schmidt's observations from Athens in the late 1800's). Schaefer (1988b, 1992a) and Doggett and Schaefer (1993) discuss the rather poor accuracy of these algorithms. A basic trouble with this approach is that different workers have interpreted the same data with greatly different conclusions (Fotheringham 1910, Maunder 1911, Ilyas 1984). This approach also has the severe trouble that the algorithm does not account for local observing conditions, so that the same criterion is applied to the humid lowlands in the east of India as well as to the high mountains in the north of India. In effect, these empirical algorithms are assuming that the whole world has the same clarity of air as Athens. A much better approach is to perform the visibility calculations as described above, so that the haziness from each site is accounted for appropriately. Bruin (1977) was the first to use this approach, but unfortunately, he made a variety of errors in his modeling that were many orders of magnitude wrong (Schaefer 1988b). A correct algorithm for predicting crescent visibility is presented in Schaefer (1988b, 1988c), and as a computer program in Schaefer (1990a).

Extensive efforts have been made to collect observations as a test of the historical criteria as well as the modern algorithm. Schaefer (1988b) has collected 201 observations from the astronomical literature and finds the ancient algorithms have a zone of uncertainty larger than the earth's circumference while the modern algorithm is 2.3 times better than all other algorithms. Doggett and Schaefer (1993) have collected 1534 observations from five "Moonwatch" campaigns, where 2522 observers spread out across North America reported whether the crescent was visible on five nights between 1987 and 1990. The conclusion was that the average error for the modern algorithm was 11° in longitude, while the empirical algorithms had an average error of 33°, the medieval algorithms had an average error of 78°, and the ancient criteria had an average error of 80°. Similarly, the maximum error for the modern algorithm was 23° in longitude, while the maximum errors were 95°, 236°, and 206° in longitude for the other three groups of algorithms. Thus, extensive sets of modern data validate the accuracy of the modern algorithm and proves it to be greatly more accurate than any other algorithm.

A related question is the age of the youngest crescent ever seen. Schaefer, Ahmad, and Doggett (1993) point out that many of the early claimed records were actually recorded long after the observation itself and the recorded weather conditions for the purported time of observation was rain, so that the apparent record is just a matter of an error in date. The current record for the earliest crescent sighting is 15.4 hours for unaided viewing by Julius Schmidt and 13^h28^m for a sighting with binoculars by Robert C. Victor.

If the moon were an ideal sphere, then the crescent should appear to extend from pole to pole, for a total angle of 180° subtended around the center of the lunar disk. Danjon (1932 and 1936) shows that the actual angle is considerably less than 180° when the moon is close to the sun. Schaefer (1991f) explains the shortening as a natural consequence of the crescent's rapid brightness decrease

towards the cusps, such that the threshold for visibility to the eye is reached at angles less than 180°. The model is shown to closely match all of the data of Danjon as well as of the Moonwatch observers.

3.12 Earthshine

On Earth, moonlight from a full Moon can brilliantly illuminate a night time landscape. Similarly, on the Moon, earthlight from a full Earth can shine brightly over a moonscape. The full Earth is much brighter than the full Moon since the Earth is so much larger and more reflective. So at times near new Moon (when the Earth is full and the glare from the sunlit lunar hemisphere is minimal), we can see the Earthlit moonscape as a ghostly orb. Leonardo da Vinci was the first to propose the correct explanation for this "Earthshine."

During a total solar eclipse, Schaefer (1991b) has calculated that the Earthshine should be barely visible despite the glare from the solar corona and multiply scattered sunlight. Peters (1970) and Durst (1982) present pictures in which lunar mare are visible by Earthshine (at least in the original photos).

3.13 Moon illusion

The rising Moon is perceived to have a ~3 times larger size than when the Moon is high in the sky. The "Moon illusion" also applies to the Sun and star groups like the Pleiades. This effect is one of the strongest and best known optical illusions, with discussions as far back as Aristotle and Ptolemy. The fact that it is an illusion can be easily proved with careful observations comparing the high and low Moon to small disks along a yardstick held up to the eye. The best discussion of the Moon illusion that I know of is Minnaert (1954), although Hershenson (1989) gives an extensive and up-to-date treatment. Many explanations have been proposed (Rees 1986), but there is no consensus as to the true cause of the Moon illusion. The perceived size is tied up with a variety of unconscious cues used by our brain to ascribe a distance to the Moon. These cues include the know distance and size of horizon objects near the Moon, the angle from the body that the eyes look to see the Moon, the eye's fixation distance, and the presence of clouds and haze. It is likely that more than one effect contributes to the overall illusion.

Despite the large perceived size of the Moon near the horizon, there are actually three effects that can make the Moon smaller near the horizon. The first effect is differential refraction, which shrinks the Moon in the vertical direction only when it is near the horizon. The second effect is from the fact that the Moon overhead is closer than when it is near the horizon by the radius of the Earth. The third effect is that the Moon's distance from the Earth varies, so that the horizon Moon at apogee may be compared to a memory of the zenith Moon at perigee. These effects typically result in areal enlargements of 16%, 3.5%, and 30% respectively.

3.14 Lunar eclipses

During a lunar eclipse, the Moon serves as a screen on which the shadow of the Earth is cast. The brightness within the penumbra (partially shadowed region) is accurately given by the geometrical fraction of the Sun covered by the Earth. The brightness in the umbra (totally shadowed region) is complicated to calculate since the only sunlight that can reach the Moon must be refracted through the Earth's atmosphere. Link (1969) gives the definitive treatment of the eclipse brightness calculations. The umbral brightness of lunar eclipses varies dramatically from one eclipse to the next, and there may even be inhomogeneities within the Earth's shadow (*e.g.*, O'Meara 1993). These variations are caused by changes in the average clarity of our atmosphere, primarily changing due to volcanic eruptions placing a large amount of aerosols in the stratosphere. Keen (1983) collected brightness estimates from 21 lunar eclipses and found strong correlation with volcanic activity. His median m_v of the Moon at mid-eclipse is *-0.9*, although the values range from *+4.1* to *-3.0*.

The color of the totally eclipsed Moon is "brick red" because only red light can pass through the long path lengths in the Earth's atmosphere when it is refracted into the umbra. This coloration can be detected with the unaided human eye only when the umbra covers more than half the Moon, since otherwise glare from the bright penumbral regions will swamp the faint red umbral light (Schaefer 1991c). No coloration is ever detectable within the penumbra, since even a thin slice of direct white solar light is orders of magnitude brighter than the red refracted light.

When an eclipsed moon is low on the horizon, the Sun must also be near the horizon so that the high sky brightness and the large extinction may combine to render the moon invisible. Schaefer (1990c) presents a theoretical and observational discussion on the visibility of low altitude lunar eclipses. Few observations have been made to test the theoretical calculations, and it appears that the theory may be optimistic by perhaps a degree of altitude.

3.15 Appulses, double stars, and occultations

Whenever a star or planet is close to a bright source, it may be hidden by the glare. This case can arise in many circumstances, including Venus' conjunctions with the Sun, lunar appulses with stars or planets, planetary satellites near the planet, and planetary appulses. A complete analysis of visibility under these circumstances is presented in Schaefer (1991c). For visibility with the unaided normal eye, the limiting magnitude is

$$m_{lim} = 4.01 - 5\,log\left(1 + 0.00372\,\theta^{-1}\,dex\left[-0.2\,m_v\right]\right) \quad , \tag{50a}$$

where m_v is the visual magnitude of the glare source and θ is the angle between the two sources. This threshold is derived from Equations 8a, 9, 10, 16, and 34 together with the assumption that $kX = 0.3$ and that the source is close enough so that day vision is involved. If the glare source is hidden from direct view, then

$$m_{lim} = 4.01 - 5\log\left[1 + 0.00184\,\theta^{-1}\,dex(-0.2\,m_v)\right] \quad . \tag{50b}$$

For cases where night vision is involved (when the glare source is faint or θ is large), the coefficients in the equations will change.

For cases where optical assistance is used, the expressions take on more complicated (but easily derived) forms. A particular case of prominence is the visibility of double stars. For a double to be split, the separation must be greater than the resolution and the fainter star must be bright enough to be seen against the background. The first condition was discussed previously, and the second condition yields limits similar to the above equations (*cf.* Schaefer 1991c).

Lunar occultations are an important and complex case of lunar appulses. Schaefer, Bulder, and Bourgeois (1992) derive the visibility of lunar occultations and confirm the model with data from 1739 observed occultations. Schaefer (1993a) implements this model into a computer algorithm that has been adopted with minor modifications by the International Occultation Timing Association.

3.16 Twinkling

Under a clear sky, the twinkling of stars creates an atmosphere of liveliness. This rapid change in a star's apparent brightness is termed scintillation. Even though stars subtend infinitesimal solid angles, they appear in a telescope as a finite disk with fuzzy edges. This image blurring is called "seeing." When a star is viewed through a small telescope, the light appears to move around like a will-o'-the-wisp dancing around a fairy. This effect is called image movement. All three phenomena are closely related manifestations of turbulence in the atmosphere. The correct idea was first advanced by Robert Hooke in 1665 when he suggested the existence of "small, moving regions of atmosphere having different refracting powers which act like lenses." The refractive index of the air varies slightly from point-to-point due to small changes in temperature and density caused by turbulent motions of the winds and heating from the ground. So the path of a beam of light passing through the atmosphere will be bent and kinked from the random scatterings imposed by the weak refractive prisms of air. An observer on the ground will be able to see light from a point source by looking in many directions at once. This spreading of the light into a "seeing disk" is caused by many small angle scatterings, and hence (by the Central Limit Theorem) has a two-dimensional Gaussian distribution. As the wind blows the eddies across the line of sight, the number and centroid of paths will shift randomly resulting in scintillation and image movement.

The last two decades have seen a tremendous rise of interest in seeing and scintillation phenomena. The motives behind the enhanced interest have included the need to improve the seeing at astronomical observatories, the need to transmit laser beams through the atmosphere for communications and military applications, and the need to optimize the resolution of photographs taken from aircraft and satellites. The best introductory article on this topic is Young (1971), while Mikesell, Hoag, and Hall (1951) provide a good discussion on the observational properties of

scintillation. A review paper by Coulman (1985) gives a detailed technical discussion along with an extensive bibliography.

3.17 Extinction angle

The extinction angle is the altitude above the ideal horizon that a star first becomes visible as it rises at night. A detailed model and computer program for calculating the extinction angle, as well as reports on observations from two sites which verify the model predictions, have been presented by Schaefer (1986 and 1987b). As an illustration of the basic calculation, an approximate value for the extinction angle for Aldebaran, as viewed from Medicine Mountain, WY (altitude 2940 m, latitude 44.8°N, and relative humidity of near 50%), in the summer will be determined, as follows: the star's m_v-value is close to *0.85 mag* and the summer k_v is around *0.14 magnitudes per air mass* (see Equations 4 through 7). For a dark sky and a normal observer, let us use the canonical limit of sixth magnitude, so that Aldebaran will suffer *5.15 magnitudes* of dimming at the extinction angle. Then by Equations 8a and 2b, the extinction angle has $X = 36$ which corresponds to an altitude of *0.2°*.

However, several simplifications have been used in the example which should be removed to give the best answer. First, the rough calculation used the value of k appropriate for day vision, whereas night vision has a peak sensitivity at 0.51 microns. This increases k_R by a factor *1.35* (see Equation 4a), decreases k_{oz} by a factor of roughly *0.3* (see discussion after Equation 5b), and increases k_a by a factor of *1.10* (see Equation 6a). So for our example, (the k-values should be $k_R = 0.101$, $k_{oz} = 0.010$, $k_a = 0.041$). Second, because the observation has been (or will be made) made so close to the horizon, Equation 8b should be used instead of Equation 8a. The adjusted expression becomes:

$$\Delta m = 0.101\, X_e(8.2\ km) + 0.010\, X_L(20\ km) + 0.041\, X_e(1.5\ km) \quad , \tag{51}$$

where the air masses are evaluated as in Equation 3. Third, the sky brightness near the horizon will be different from that at zenith. Unfortunately, Equation 15a must break down close to the horizon since the sky never appears perfectly black at the horizon, but instead has a brightness comparable to that 10° higher (Pilachowski *et al.* 1989). Thus, with $B_0 = 180\ nL$, an effective $Z = 80°$, and $k = 0.152$, the sky brightness is 220 nL, so that the limiting magnitude-value will be 5.80 (see Equations 10 and 34). Hence, the extinction angle will be the altitude at which Aldebaran is dimmed by 4.95 magnitudes; from Equations 3 and 51, this occurs when Aldebaran is 0.6° above the horizon.

3.18 Telescopic limits

The limiting magnitude for stars visible through a telescope depends on many conditions. The telescope's aperture, the magnification, and the sky brightness are the primary determinants for the limits of vision through a telescope. Schaefer (1990b) presents a complete model for predicting the limit based on equations like those given in this review. The model was tested against 314 observations from a large number of observers over a wide variety of conditions, and the model was

found to be accurate and unbiased, yet with a half-magnitude uncertainty. Gaskell (1993) has demonstrated that much of this scatter was actually due to poor input data for the zenith limiting magnitude. Schaefer (1989b) presents a computer algorithm that incorporates this model.

3.19 Meteors and optical flashes

The study of meteors has been made primarily with visual observations, so that any interpretations must involve a knowledge of how the human eye/brain combination detects moving flashes of light. Important questions include "How to correct an observed rate to standard conditions?," "What is the effective field-of-view as a function of magnitude?," and "How to derive a true brightness from the perceived brightness?" Few relevant physiological studies have been published (but see Niven and Brown 1944). So the important studies have all been empirical in nature, often involving large numbers of observers. The results are well reported in Lovell (1954).

The night sky is full of flashes. Many of these are of real astrophysical origin far from the Earth. A dozen categories of stars flash with amplitudes greater than a few tenths of a magnitude and with durations of under a few minutes, including flare stars, symbiotic stars, Mira stars, and even "normal" stars (Schaefer 1989c, 1991a). Only flashes from Gamma-Ray Bursts produce events bright enough to be detected with unaided human vision, although these occur at the rate of only several per year and last for only a few seconds (Hudec 1993, Schaefer 1985a). Optical transients from Gamma-Ray Bursts are of such importance that many groups around the world have started multi-million dollar experiments designed to detect them (see articles in Paciesas and Fishman 1992 and Ho, Epstein, and Fenimore 1992) despite grave difficulties. These problems include the rarity of the flashes, the lack of knowledge of where or when to look, and the high rate of near-Earth flashers. Foreground sources include airplanes, meteors, and satellite glints, while many phenomena may mimic a flash for some detection programs (Schaefer 1985a and Schaefer *et al.* 1987b).

4. Applications

The tools discussed in the previous section are often ends in themselves, in that they answer questions raised about celestial visibility. For example, the mathematical model of Codona provides an explanation for the shadow bands during a solar eclipse, an evaluation of Equation 49 will yield the threshold for discovering a sun grazing comet in the daytime, and the optimal conditions for viewing the green flash may be derived from the work of Menat. But for many applications, the questions can be answered only by detailed examination of particular circumstances. That is, the tools from the previous section apply to general cases, but this section concerns the application of these tools to specific cases.

The applications of celestial visibility are extremely broad, so this section cannot hope to do more than touch on some of the more important or interesting topics. I have selected six topics with several applications that typify the breadth of the field. Unfortunately, the review on each topic must be brief, even though virtually every sentence in this section has been the subject of more than one

article or book. Nevertheless, as appropriate for a review, I will present the problem, outline the solutions, and identify the key references.

4.1 Archeoastronomy

The primary paradigm of archeoastronomy is that the designs of archeological sites have alignments with the rising and setting of celestial sources. Although there had been earlier work on the topic, archeoastronomy first became widely known with the claim by Hawkins (1965) that Stonehenge was a sophisticated astronomical observatory build to predict eclipses and incorporating solar and lunar sightlines. The other archetypal study is that of Thom (1971) in which British megalithic sites were identified as highly accurate lunar observatories. These two influential claims have inspired a vast controversy and a worldwide search for similar alignments.

The archeoastronomy paradigm requires a calculation relating the archeologically determined azimuth with some celestial object's declination. This azimuth-to-declination conversion has mandatory corrections for refraction, parallax, and altitude of first visibility. The average refraction correction is well known, but the size and implications of the variations about this average have only recently been learned (Schaefer and Liller 1990, Weiss 1990). For low horizon events at sites away from the equator, the uncertainty in the correction translates into a limit on the accuracy of the archeoastronomy paradigm. Thus, the few claims which rely on high accuracy (chiefly those of Thom 1971) are no longer tenable. The variation of the parallax correction introduces additional uncertainty for lunar sightlines. The altitude correction will depend on the type of source and alignment. For sunrise or moonrise, there is always a basic uncertainty (for cultures with no ethnographic background) as to whether the time of first gleam, bisection of the disk, or complete visibility was used. For star rise, both the altitude of first visibility and the variation of the extinction angle have been greatly underestimated in most published material (Schaefer 1986). Similarly, for alignments to the heliacal rising of stars, the altitudes used in most of the literature are lower than valid for modern theory or observation (Schaefer 1987a). The end result of all these variations and systematic errors is that high accuracy alignments are impossible and that many proposed alignments don't point as claimed.

Archeoastronomy accepts alignments towards the rising or setting of the Sun at the solstices, the Moon at its extreme declinations (also referred to as "lunastices," or "lunar standstills"), the brighter stars, and the cardinal directions as being of possible significance. In the absence of ethnographic evidence to winnow this list, there are roughly two dozen directions along the horizon which might be considered important. If the indicated azimuth need be within 1° of the calculated azimuth, then any sightline has a roughly 13% chance of being taken as intentional. Therefore, an alignment at a one-axis monument cannot be regarded as statistically significant, while most monuments have many reasonable lines such that we would expect many apparent alignments by coincidence.

Modern archeoastronomy has advanced greatly since the early days of alignment searches. First, realistic statistical studies have been performed, with the best example being a series of papers

by Ruggles (1981, 1982, 1983) wherein the British megalithic lunar observatory hypothesis is shown to be insignificant. Second, the emphasis has turned to low precision alignments where the examination of many sites can yield significant results (*e.g.*, Ruggles 1984). Third, archeoastronomers are paying attention to the constraints and backgrounds deduced from archeology (*e.g.*, Serio, Hoskin, and Ventura 1992). Fourth, the motivations and interpretations are being based on ethnographic evidence. An excellent summary of current work is given in Aveni (1989), where archeoastronomy is revealed as a mature and believable discipline.

4.2 Christian issues

The rationalizing impulse of modern man has frequently turned to questions relating to the history of religion. The goal is usually to provide an explanation for a reported miraculous event that only invokes natural phenomena. Although the purpose of these apologetics is rarely stated, I suspect that the primary idea is to buttress the faith of believers in the historical validity of the sacred writings. Unfortunately, such attempts must remain as an intellectual game since a believer can accept the miraculous event either as a true miracle or as some natural phenomenon prepared by the deity. Alternatively, a nonbeliever can always accept the everyday explanation without requiring a deity or indeed can merely reject the historical validity of the account. Celestial visibility figures large in these *apologia*, since the sky is commonly the stage for the miraculous and because the positions of stars and planets in ancient times can be reliably reconstructed.

Christianity has a rich history of events and miracles that relate to the sky, and many of these have been provided with explanations in terms of celestial visibility. Thus, Jacob's ladder may have been crepuscular rays, Ezekial's vision may have been a halo complex, the parting of the Red Sea and the drowning of the Egyptian army might have been caused by inferior mirages, as might the walking on water by Jesus (Fraser 1975). Alternatively, the parting of the Red Sea and the subsequent drowning of the Egyptian army (as well as the plague of darkness for three days) have been related to the tsunami and ashcloud from the titanic eruption of the volcano on Thera. Halo displays could have caused crosses seen in the sky that served as the inspirations for both the conversion of the Emperor Constantine and the design of the Danish flag. The final split between the Eastern Orthodox and Catholic Churches occurred a few days after the appearance of the supernova of 1054 AD, with a declaration that God had given a sign in the sky (O. de Jager 1991; private communication). At the fall of Constantinople, the departure of the Holy Ghost was indicated by a lunar eclipse and a strange light display over the Church of the Holy Wisdom (Runciman 1965).

The Star of Bethlehem is the best known example of a Biblical event identified with prosaic astronomical causes. Unfortunately, so little information is presented in the Gospels that many reasonable theories are possible. The Star has been identified with a comet (Humphreys 1991), a nova (Clark, Parkinson, and Stephenson 1977), a triple conjunction between Jupiter and Saturn (Hughes 1976), a massing of Mars, Jupiter, and Saturn in Pisces, an occultation of Jupiter by Venus (Sinnott 1986), a supernova (Morehouse 1978), and the stationary point of Jupiter (Bulmer-Thomas 1992). Extensive discussions from an astronomical/chronological point of view are in Finegan (1964) and

Vardaman and Yamauchi (1989). Perhaps the best conclusion with regards to this contentious question is that the nature of the Star of Bethlehem is unknown and that Jesus was born shortly before Herod's death in 4 BC (as dated by a lunar eclipse associated with the latter's death).

On the other hand, the Crucifixion is a chronological problem for which there is almost enough information to deduce the exact date (Fotheringham 1924, Finegan 1964, and Vardaman and Yamauchi 1989). The traditional statement of the task is to find a day which is a Friday on either the fourteenth or fifteenth of the Jewish month of Nisan in the years AD 26-36 (preferably from roughly AD 30-33). Nevertheless, a minority of researchers seek to change some aspect of the statement, with perhaps the largest group being those who claim that the Crucifixion happened on a Wednesday (Scroggie 1948). The required calculations are astronomical, since the days within a lunar month can only be placed on a modern calendar by means of crescent visibility determinations (Schaefer 1990c and references therein). After these calculations are made, the only possible dates are 7 April AD 30 and 3 April AD 33. Most writers have chosen between these two alternatives on fairly weak grounds, the best known of which is the fact that a lunar eclipse occurred on the later date. The connection of this eclipse to the death of Jesus is made when Peter talks about the Moon turning into blood four verses before a brief mention of the Crucifixion. However, such a connection is likely to be wrong since the verse discussing the possible lunar eclipse explicitly refers to the Judgement Day. In any case, a slight umbral eclipse cannot color the Moon like blood (Schaefer 1991c) and the lunar eclipse of 3 April AD 33 was unlikely to be visible from Jerusalem (Schaefer 1990c).

4.3 Islamic issues

The Night of Power is the night on which the prophet Muhammad received the first revelation of the Qur'an from the archangel Gabriel. Modern Muslims take its anniversary to be especially blessed, although the exact date is not known. From the traditions, it apparently occurred on an odd-numbered night in the last ten days of Ramadan around the year 610. Hawkins (1978) proposed that the date was 23 July 610, during which a crescent/Venus conjunction and the surrounding sky might match several verses of description in the Qur'an. However, this thesis can be ruled out since the proposed date was an even-numbered day in Ramadan and because it was a Thursday while the Night of Power was a Monday (Schaefer 1991e).

Two of the primary responsibilities of any Muslim are to make the five daily prayers and to perform a pilgrimage to Mecca. Both of these duties must be performed at certain times in relation to astronomical events. The prayer times are defined by the apparent altitude of the Sun, with full details in Ilyas (1984). The start of dawn and the end of dusk is often taken to be when the Sun is 18° below the horizon, although variations by up to 2° have been adopted (Ilyas 1984, King 1975). The possibility of variable refraction might require the expansion of the excluded times around sunset (Weiss 1990). The correct prayer times at high latitude and in space are not well defined.

The timing of the pilgrimage, and indeed the start of each Islamic month, is based on the visibility of the first crescent after new Moon. However, many different criteria have arisen to handle cloudy weather and the verification of sightings. The rapid communication between Islamic

communities has caused discord since holy days can be celebrated up to two or more days apart. Also disconcerting is that this rapid communication allows one alleged sighting in a distant land to start the month for many groups throughout the world, even though the sighting was made before conjunction, so by definition the sighting could only have been of an old Moon (Ilyas 1984, Ahmad 1992). In recent years, the majority of Ramadans have been started based on claimed detections before the conjunction (Ilyas 1984, Ahmad 1992). These early "sightings" are a simple consequence of having a large number of honest lay observers looking for the Moon with high anticipation, so that the normal false alarm rate (measured to be as high as 15% by Doggett and Schaefer 1993) will assure a crescent "detection" on any evening on which many people look.

The Ka'ba is an ancient building in the Great Mosque, Al-Haram, in Mecca and is the most holy building of the Islamic faith. Several astronomical associations concerning the Ka'ba have been advanced. Thomsen (1980) has made a reasonable case that the Black Stone, Al-Hadjar Al-Aswad, consists of impactite glass derived from the meteoritic craters of Al-Hadidah, called Wabar. Hawkins and King (1982) discuss several medieval texts as well as modern plans of the orientation of the Ka'ba and suggest that the building was roughly aligned to the southernmost setting of the Moon and to the rising points of Canopus, the Handle of the Big Dipper, and Antares. Both mosques and daily prayers should be directed towards the Ka'ba, a direction known as the the Qibla. This alignment is ultimately an astronomical problem, which Ilyas (1984) describes in detail. King (1980 and 1982) discusses the Qibla and mosque orientations from a historical point of view. The siege of the Great Mosque in Mecca in November 1979 by Islamic fundamentalists was timed to coincide with the start of the new lunar century when tradition has it that the Mahdi (a messianic deliverer of the Shi'i sect) will restore the true religion before the Judgement Day.

In modern times, the star and crescent symbol is associated with the Islamic religion. The astronomical nature of the icon suggests that it might have its origin in some historical conjunction of the Moon with a star or planet. Indeed, legends attribute the origin to a crescent/Venus conjunction on the night of the Battle of Sakarya on 26 August 1922 in Turkey, to either a crescent/Venus conjunction or a crescent alone or a lunar eclipse around the fall of Constantinople in 1453, to a crescent/Venus conjunction on the Night of Power in 610, or to a celestial light that saved Byzantium during a siege by Philip of Macedon. Unfortunately, each of these legends cannot be reconciled with what was actually visible in the sky (Schaefer 1991e). Many non-astronomical origins have also been advanced, including a dream of the Sultan Othman, a stirrup decoration, an ancient emblem of Bosnia, an ancient Illyrian device, the symbol of Diana and the Virgin Mary, or two bear claws. But instead of testing each of these proposals individually, it is more profitable to track the actual usage of the star and crescent back in time. Ettinghausen (1971) does a definitive job of tracing it back to the year 651, where the earliest Islamic usage is on a coin with eleven star/crescent pairs which is virtually an exact copy of a Sassanian series which first appeared around 490. So the star and crescent symbol was not originally an Islamic symbol, but was only borrowed. In fact, the symbol can be traced on coins, medallions, inscriptions, stellae, and cylinder seals throughout the Middle East as far back as *c.* 2500 BC (Schaefer 1991e). This immediately disproves all of the above suggestions as the true

origin. So it appears that the event or symbolism which motivated the modern icon is lost in the pre-historic ages.

4.4 Supernovas

The study of supernovas is one of the most important areas in all modern astronomy since there are important implications for virtually all other astronomical disciplines. Historical observations of galactic supernovas have provided the fundamental parameter of age for the associated remnant. The definitive analysis is in Clark and Stephenson (1977), although further work has suggested additions and modifications to this work. For example, Thorsett (1992) has argued that their identification for the remnant of the supernova of AD 185 cannot be correct because the heliacal rise dates would not match the reported times of visibility. Thorsett then makes an attractive suggestion that the remnant is actually PSR1509-58 (in supernova remnant MSH15-52) a pulsar with a spindown age of 1690 years. However, the heliacal rise dates for his candidate can match the reports only if the supernova was -4 magnitudes or brighter. Given the distance and extinction to the pulsar, this implies a peak absolute magnitude of at least -20.2, which is brighter than all known Type II supernovas. (The explosion had to have been a Type II in Thorsett's identification since a pulsar was left behind.) It might be possible to reconcile the visibility limit with Thorsett's identification if a small value of Hubble's Constant is adopted, the supernova were a Type IIP, and the event equaled the previous most luminous known explosion.

Brecher (1987) has pointed out that there is reasonable historical evidence that supernovas can rebrighten significantly within several decades after the original explosion. Ashworth (1980) makes a reasonable case that Flamsteed recorded the supernova that formed Cas A. De Vaucouleurs and Corwin (1985) review the visual observations of the 1885 supernova in the Andromeda Galaxy.

4.5 Eclipses

In olden times, eclipses were often viewed as miraculous portents, and so were frequently recorded in many chronicles. A study of these eclipse reports and the associated eclipse visibility can be used to derive information of historical value. In general, the greatest utility comes from relating historical events with the modern calendar. For example, the death of King Herod is tied to a lunar eclipse in 4 BC (Hughes 1979, but see Edwards 1972) and this serves as the primary constraint on the birth of Jesus and the identification of the Star of Bethlehem. The absolute chronologies of most old civilizations with written records are based on the known dates of astronomical events, primarily eclipses. The work on this problem is widely scattered in the literature, but Newton (1972) offers an excellent practical treatise for medieval European records. There are numerous instances of eclipses affecting historical events, but other than in some rare cases (the Crucifixion eclipse and General Charles G. "Chinese" Gordon's lunar eclipse) the details of visibility do not affect the historical interpretation (Schaefer 1992c).

The problem can be turned around so as to derive astronomical information from historical eclipse reports. The primary astronomical question to be answered relates to the acceleration of the rotation of the Earth due to various incompletely understood tidal forces. Slight changes in the length of the day will accumulate over the centuries so that the orientation of the globe will be significantly different than if the day was unchanging. These differences will only be noticeable when the position of planetary bodies can be accurately related to a location on the Earth's surface, such as during an eclipse. The idea is that a reliable ancient report of a specific total solar eclipse from a known site will allow a determination of the difference between Universal Time (a measure of the Earth's orientation) and Dynamical Time (a uniform measure of time). With enough reports throughout the centuries, the time derivative of these differences can yield the tidal forces on the Earth. The best references on this topic are Newton (1972, 1979) and Stephenson and Morrison (1984).

4.6 Sunspots

The sunspot count is the premier measure of solar activity, in part because a record is available for the last two millennia. But its reliability must be evaluated before this data can be used. Hoyt and Schatten (1992) and Schaefer (1993b) have concluded that the formulation of the sunspot count that uses individual spot numbers are unreliable and subject to drifts, whereas a simple group count is unbiased. For pre-telescopic observations, Schaefer (1991d) shows that systematic biases due to observer age, diet, and filtering technique should not exist. Stephenson (1990) attempts to use the frequency of observations of happenings in the sky to normalize for the variable effects of interest in recording celestial events and of the preservation of the accounts.

The sunspot record is tabulated in Waldmeier (1961) for telescopic observations and in Yau and Stephenson (1988) for pre-telescopic observations. The measure of proxy indicators of solar activity (*e.g.*, auroras and tree ring widths) are summarized in Schove (1983). Excellent summaries of the correlations between the sunspot and climatic indicators are in Eddy (1980) and Sonnett, Giampapa, and Mathews (1991).

5. Observational projects

Celestial visibility often tackles problems involving naked eye astronomy. As such, it would seem that most of these problems should have been solved by the ancients or at least back in the nineteenth century, but this is not true. As an example, the modern analysis of heliacal rise phenomena in Schaefer (1987a) represents the first useful report of observations since the Babylonians and the first theoretical advance since Ptolemy. The reason is that the ancients were rarely quantitative and usually did not even think of the questions of interest to current researchers. Even if old-time astronomers had the right frame of mind, the necessary tools to create a realistic model for celestial visibility were not there. And then in modern times, astrophysics has made many impressive advances for which visual astronomy has no contribution. I can hardly fault today's scientists for trying to solve the enigma of gamma-ray bursts while the orientation technique for the Great Pyramid remains unknown.

Nevertheless, as interesting as modern astrophysics is, there are still many fun and exciting questions for celestial visibility, and these questions are often of interest to a much broader community.

Studies of celestial visibility usually require little if any equipment. Most of the observational results reported in this review have only involved careful visual work, while a few needed an accurate clock or a small telescope. Indeed, it is still possible to publish papers in major journals based on naked-eye backyard observations with relevance to modern astrophysics (Schaefer 1985a, 1991c, 1993b, Schaefer *et al.* 1987a, and Schaefer, Bulder, and Bourgeois 1992). The simplicity of observational studies means that any careful researcher can make original contributions to the field, even if they do not have access to large telescopes or sophisticated electronic detectors. The simplicity also implies that celestial visibility can be used by teachers as excellent class observing projects.

The remainder of this section will be devoted to a dozen proposals for observing programs. The projects are roughly ordered by difficulty, with the last four requiring major efforts on difficult topics. Each program could lead to publishable results if a careful and comprehensive study is undertaken.

(1) What is the frequency of the green flash? Traditionally, the green flash is a rare phenomenon, but many sources report typical rates of 20% (Schaefer 1992b and references therein). Apparently the wide variations in reported rates depend on the haziness and horizon characteristics, with clean air and low straight horizons producing many flashes. It would be interesting to measure the green flash occurence frequency from many locations? Does the color of the setting Sun or the visibility distance measured at a nearby airport correlate with the flash probability?

(2) What is the variation of refraction on the horizon? This is a question of importance for many diverse communities (Weiss 1990). This variation can be measured by the remarkably simple expedient of timing the sunset over a water horizon. The idea is that the true zenith distance of the upper limb of the Sun for the exact time of observation will differ from 90° (plus a horizon dip correction for your altitude above the water) by the horizon refraction. Therefore, a series of sunset timings can be translated into a distribution of how the light bending varies (see Schaefer and Liller 1990 for details and Sampson 1993 for a sample project report). This program is of current research interest because the variation has only been measured from eight sites, and the defenders of the various precise archeoastronomical alignments have adopted the excuse that each proposed alignment is at a location for which the variation could have been much smaller than for these eight sites. To test the probability that locations can have highly regular refraction, we need experimental measures from many places.

(3) What is the probability for detecting a star near the threshold? That is, detection is certain for bright stars, detection is impossible for very faint stars, and the threshold is defined to be the magnitude where 50% of the stars of that brightness are seen. Stars near the threshold will have some detection probability that depends on how close their brightness is to the threshold. The project is to observationally define the probability as a function of magnitude. The procedure would be to map some region near zenith from a dark site, so that all stars down to some appropriate level can be labelled as seen or not-seen. This procedure will have to be repeated under similar circumstances to build up good statistics. The threshold itself is not of general interest since this depends on local

conditions, but the size of the range over which the visibility is neither certain nor impossible is of interest for various model comparisons (see Schaefer 1990b and Schaefer, Bulder, and Bourgeois 1992). No such observational studies of stars have appeared in print. This study could be repeated with many observers and for binocular observations. Physiological studies in the laboratory (Blackwell 1946) suggest that the magnitude range from 20% to 80% detectability is 1.1 magnitudes.

(4) What is the distribution for limiting magnitudes in a group of people? Some people can see much fainter than others. How does the threshold vary with age, Snellen ratio, glasses, duration of observing, or experience? This has never been measured, but it is easy to do. Just take a group of friends or a class out under a uniform sky, get them all dark adapted, then ask each person to independently map all stars in some unknown region. After some tedious bookwork, a limiting magnitude for each individual can be deduced, and then the distribution of limiting magnitudes. The results from a comprehensive study should be published.

(5) How much glare can hide a double star from view? The duplicity of a star can remain hidden either because the two stars are too close or because the glare from the primary masks the secondary. Classic examples of these two masking mechanisms for small telescopes are the ϵ *Lyræ* and *Sirius* systems. The resolution limits on splitting a double are well known (Dawes limit), but the effects of glare are not well measured. Schaefer (1991c) presents a model and a few observations on the limits of visibility of faint secondaries that are well separated from the primaries. This model needs comprehensive testing. An ideal test would be to use a wide range of aperture stops on a medium size telescope placed by an assistant in a random manner and then record through which apertures the secondary can be detected. After repeated trials, several observers, and many test double stars, a reliable data base can be created for evaluating the theoretical model.

(6) How could the Great Pyramid be oriented north/south to better than 5'? This cardinal alignment could only have been done with astronomical observations, and indeed many techniques have been proposed, but no modern worker has ever tried to actually lay out a long line to an accuracy of better than 5' with old methods. Often, harsh realities can disprove an idea that looks good theoretically or can suggest techniques that were previously overlooked by calculators with no practical experience. Indeed, my trials with the Washington Monument indicate that Neugebauer's method (Neugebauer 1980) can be greatly improved if the observer looks directly for the position with the first glint of visible sunlight as opposed to looking at penumbral shadings on the ground. I suggest that serious researchers should actually try the various techniques to see what works and what does not work. I also suggest that it would be a good class project to try to lay out a 50 foot north/south line to an accuracy of 0.1°. This class project could be approached with techniques that the ancient Egyptians might plausibly have known, with modern tools, or with both.

(7) Where was Columbus' landfall in the New World? The traditional answer is the island of San Salvador (previously called Watling's Island) in the Bahamas (Morison 1942). However, many other landfalls have been proposed over the years (De Vorsey and Parker 1985). Unfortunately, all proposed routes have some details directly at odds with Columbus' diary, while many have reasonably good agreement with the diary on most points of comparison. After much reading about this contentious topic, my feeling is that the islands of Plana Cays (Pickering 1992), Samana Cay (Judge

1986), San Salvador (Morison 1942), and Conception Island (Mitchell 1992) all have roughly equal chances of being the landfall, with no expectation of breaking this four-way tie unless new evidence is discovered. One way to distinguish the various paths is to compare the inter-island visibility reported in Columbus' diary with those along each proposed path. For example, on the afternoon of 15 October 1492 while sailing west along the shore of the second island, Columbus sees another island farther west. But Long Island (everyone's third island) cannot be seen from either Crooked Island or Rum Cay. Similar problems arise in diary entries from the crossing between each of the islands. With the help of Columbus' diary, modern maps, and the terrestrial refraction in Table 2, it may be possible to distinguish between the various routes. But don't expect to change the minds of any of the advocates!

(8) Did Admiral Peary reach the North Pole? Peary's claim has been widely accepted. Unfortunately, Peary did not write a continuous diary narration, did not take any reliable witnesses/navigators along with him on his journey, did not measure the depth of the ocean floor, did not record the magnetic deviation, nor provided reliable sextant observations near the pole. So in essence, all we have is Peary's good word as the premier arctic explorer of all time. However, old persistent doubts have resurfaced recently (Rawlins 1973, Herbert 1988) with demonstrations that no polar explorer has ever travelled anywhere near as fast as Peary claimed to have after leaving the last reliable witness (with his speed before this departure being quite normal), the impossibility of navigating a straight line to the pole without sextant measurements despite drifting of the icepack and numerous required detours caused by pressure ridges, and earlier alleged fraudulent claims by Peary. So the validity of the claim comes down to Peary's (questionable) word of honor versus the apparent impossibility of travelling fast enough and straight enough. Then, Davies (1990) claimed to have shifted the verdict by providing new evidence in the form of photogrammetric analysis of pictures taken at the northernmost camp. The idea was to deduce the height of the Sun above the horizon from analysis of shadows. If the time of the picture can be determined, then this measurement is a substitute for a sextant sighting. With several pictures spread over time, a position can be measured. Davies found Peary's pole camp to be within 20 miles of 90° North latitude (although Rawlins strongly disagrees). But how accurate can the photogrammetry be? Pickering (1990) demonstrates that the actual uncertainty region covers the entire polar cap above 85° North latitude. An optimal photograph for analysis is presented by Garrett (1990), from which it was claimed that the Sun's altitude can be determined to 0.1°. Go out and try the photogrammetric latitude measurement with other pictures, especially with pictures of scenes showing only humans and jumbled snow piles. After this exercise, does Davies' new evidence offset the modern verdict? Even if Peary did not reach the North Pole (and knew that he didn't), this cannot distract from Peary's many remarkable achievements made over a lifetime of polar exploration.

(9) Can sky color and brightness be predicted? The scattering and absorption of light by aerosols, gases, ozone, and the Earth's surface is an extremely complex problem with no analytic solution. Yet it is possible to write a Monte Carlo computer program that tracks large numbers of photons through the atmosphere while correctly treating each scattering or absorption. The details of the program are not too difficult, although the program would take a long time to run. Once many

photons have been followed, a map of the sky brightness can be constructed. By running the program for various wavelengths, it should be possible to calculate sky colors. The program could be tailored easily to run for any haziness and Sun altitude.

(10) What is the limiting magnitude for the visual detection of galaxies? This question is of constant importance to amateur astronomers who seek faint galaxies (or nebulas). The amateur community has a long and strong tradition of searching for faint "deep sky" objects (see the long running "Deep Sky Wonders" column by Walter Scott Houston in the periodical *Sky and Telescope*). This literature is full of statements about the minimum aperture and conditions needed for viewing many individual sources, but neither a mathematical formula nor even a crude rule-of-thumb has ever appeared for galaxy thresholds. (This is starkly different from the case for point source thresholds; see Schaefer 1990b and references therein.) The problem is that the physiological limits for fuzzy sources are not known. Nevertheless, it may be possible to compile published information, analyze the trends, and produce an empirical equation for nebular visibility. Or, carefully designed experiments (*cf.* Blackwell 1946) might be run to provide the needed physiological data, so that a set of equations can be constructed from first principles that will solve the problem.

(11) What is the aperture correction for cometary magnitude estimates? It is well known that cometary magnitudes measured with large telescopes are systematically fainter than those measured with small telescopes, so that some aperture correction is needed for an unbiased measure of brightness. This is an important problem in any study of the comet brightness variations, since large systematic biases will enter the data as bright comets will appear brighter and faint comets will appear fainter. Various observational and simplistic theoretical studies have appeared (Morris 1973, Öpik 1963, Bobrovnikoff 1941), although the results differ to a large degree. What is needed is a real theoretical study based on the physiology for detecting nebulous sources, supported by an empirical test of the validity of the resulting model. This will be a difficult project, as no physiological data for fuzzy sources has been published. The reward for success will be an accurate cometary aperture correction, from which unbiased cometary magnitudes can be extracted, resulting in an accurate picture of how comets brighten.

(12) How does the activity of each solar cycle vary over long periods of time? The long term activity of the Sun (as measured by the sunspot count) is directly tied to the average temperature on Earth, with a well known example being the Maunder Minimum and the "Little Ice Age" (Eddy 1983). This connection makes an understanding of the long-term solar activity very important for policy decisions in an age when small changes in the Earth's climate could mean starvation for some fraction of the many billions of inhabitants. The best record we have for long-term solar activity is the sunspot number, which goes back to the early 1600's with reasonable coverage. The method for calculating the old Zürich and modern International sunspot numbers scales the observations of all observers to those of a single "standard" observer. Unfortunately, all observers (including the "standard" observer) are subject to variations in their scaling factor over time with an *rms* scatter of typically 10% and maximum deviations of up to a factor of two. The causes of these variations include changing seeing conditions, the aging of the eyes, and the evolution of judgement on what is a spot and what is a pore. The end result could be an overall modulation of the sunspot count on all

time-scales depending on the vagaries of a single observer. Fortunately, the reasonable correlation between modern radio flux measures and the sunspot count demonstrate that the drifts are less than 20% within several decades. But for longer time intervals, the drifts could be quite large. So for example, the low cycles just before the Maunder Minimum might be due to the fact that it was just before the Maunder Minimum or due to the fact that Galileo used a tiny telescope. The current procedure for measuring the sunspot count cannot avoid the problem of random modulation and slow drifts, although Schaefer (1993b) has demonstrated a new method for measuring the *K* factor which is immune to drifts. Equivalently, a group count could be used as in Hoyt and Schatten (1992). This opens up the large and important program of re-evaluating old data to produce a revised sunspot count from 1610 to present.

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